

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\109-DA-DB-Rev 01.pvdb

Date of Analysis : Jul 30, 2026 9:29am

PV Elite 28 U2, April 2026



Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 5350.000 mm ID	
Vessel Design Length, Tangent to Tangent	290.00 cm
Distance of Bottom Tangent above Grade	790.00 cm
Distance of Base above Grade	0.00 cm
Specified Datum Line Distance	790.00 cm
Internal Design Temperature	275 °C
Internal Design Pressure	44.531 kgf/cm ²
External Design Temperature	275 °C
External Design Pressure	1.055 kgf/cm ²
Maximum Allowable Working Pressure	45 kgf/cm ²
External Max. Allowable Working Pressure	12 kgf/cm ²
Shop Test Pressure	59.927 kgf/cm ²
Field Test Pressure	57.890 kgf/cm ²
Required Minimum Design Metal Temperature	-12.0 °C
Warmest Computed Minimum Design Metal Temperature	46.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-387 11	2	...	K11789	
Head	SA-387 11	2	...	K11789	
Skirt	SA-516 70	...	...	K02700	
Skirt	SA-387 11	2	...	K11789	
Nozzle	SA-182 F11	2	...	K11572	
Nozzle Flg	SA-182 F11	2	...	K11572	
Basing	SA-36	...	...	K02600	
Base Bolting	SA-307 A	...	6 <= t <= 100		

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt-3	...	...	...	1.6000	N/A	No
Skirt-2	...	...	...	1.6000	N/A	No
Skirt-1	...	...	...	1.6000	N/A	No
Bottom head	44.531	1.05	46.000	3.0000	N/A	No
Shell	44.531	1.05	45.000	3.0000	N/A	No
Top head	44.531	1.05	46.000	3.0000	N/A	No

Element Types and Properties:

Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Skirt	-470.0	320.0	22.0	22.0	...	...	1.00	1.00



Skirt	-120.0	350.0	22.0	22.0	...	...	1.00	1.00
Skirt	0.0	120.0	22.0	22.0	...	...	0.70	0.50
Sphere	0.0	0.0	49.0	44.0	42.9	11.2	1.00	1.00
Cylinder	290.0	290.0	84.0	84.0	83.4	20.1	1.00	1.00
Sphere	290.0	0.0	49.0	44.0	42.9	11.2	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N2_600	WRC-107/537	0.971	n/a	Passed
N1_600	WRC-107/537	0.971	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support cm	Cumulative Wind Shear kgf	Earthquake Shear kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	160	3394.01	27390.6	22384.4	256929
20	30	495	2574.52	27002.3	12834.8	169900
30	40	730	1678.2	26066.3	5392.52	77030
40	50	789.999	1370.89	25593	3563.11	46035.1
50	60	935	1370.88	21094.9	3563.07	46034.4
60	70	1080	593.401	5818.69	714.877	7009.85

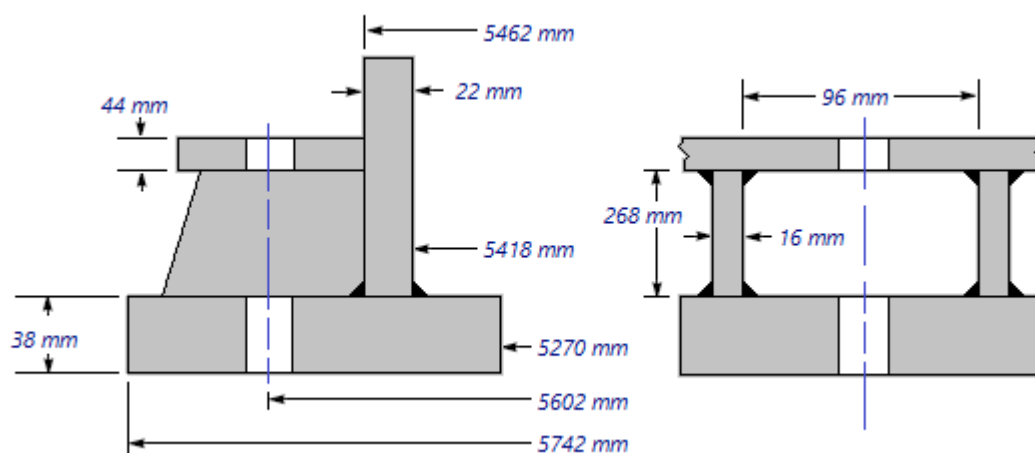
Abs Max of the all of the Stress Ratio's : 0.9638

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	38.0000	mm
Inside Diameter of Basing	5270.0000	mm
Outside Diameter of Basing	5742.0000	mm
Nominal Diameter of Bolts	48.0000	mm
Diameter of Bolt Circle	5602.0000	mm
Number of Bolts	24	

Thickness of Gusset Plates	16.0000	mm
Average Width of Gusset Plates	140.0000	mm
Height of Gussets	268.0000	mm
Distance between Gussets	96.0000	mm
Thickness of Top Plate or Ring	44.0000	mm
Radial Width of the Top Plate	140.0000	mm
Number of Gussets	48	

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View



Loads for Foundation/Support Design:

Total Wind Shear on Support	9616.	kgf
Total Earthquake Shear on Support	33613.	kgf
Wind Moment on Support	132912.	kgf-m.
Earthquake Moment on Support	367456.	kgf-m.
Maximum Initial Bolt Load	3630.	kgf

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	115269.3	kg
Shop Test - Fabricated + water (full)	262340.6	kg
Shipping - Fab. + rem. intls.+ shipping app.	160335.1	kg
Erected - Fab. + rem. intls.+ insul. (etc)	160335.1	kg
Empty - Fab. + intls. + details + wghts.	160335.1	kg
Operating - Empty + operating liquid (no ca)	174684.9	kg
Field Test - Empty weight + test medium (full)	267887.0	kg

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Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N2_600	46	OK	...	OK	12.38	OK	Passed
K3_80	46	OK	...	OK	7.80	OK	Passed
UC-1	45	OK	...	...		OK	Passed
M1_600	46	OK	...	...		OK	Passed
K1_50	46	OK	...	OK	6.42	OK	Passed
N1_600	46	OK	...	OK	12.06	OK	Passed
K2_50	46	OK	...	OK	6.42	OK	Passed
K4_80	46	OK	...	OK	7.80	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 45 Nozzle : UC-1
 Minimum MAWP Shells/Flanges : 45 Element : Shell
 Minimum MAPnc Shells/Flanges : 47 Element : Shell

Computed Vessel M.A.W.P. : 45 kgf/cm²

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
40	N2_600	0.000	0.000	1132.000
40	K3_80	0.000	0.000	211.431
50	UC-1	900.000	0.000	1212.000
60	M1_600	0.000	0.000	1022.907
60	K1_50	0.000	240.000	182.945
60	N1_600	0.000	0.000	1122.920
60	K2_50	0.000	120.000	182.945
60	K4_80	0.000	180.000	170.900

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
K1_50	50 mm	XXS	WNF	60.3	11.074	...	...	1087	300
K2_50	2 in	XXS	WNF	50.8	0.000	...	...	387	300
K3_80	80 mm	160	WNF	88.9	11.125	...	...	597	300
K4_80	80 mm	XXS	WNF	88.9	15.240	...	...	2159	300
N2_600	600 mm	Actual	WNF	600.0	20.000	...	...	644	300
UC-1	600 mm	Actual	LWN	640.0	20.000	...	...	473	300
M1_600	600 mm	160	WNF	609.6	59.512	...	...	1393	300
N1_600	600 mm	80	WNF	609.6	30.937	...	...	1076	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K1_50	SA-182 F11	44.000	10.000	...	...	...
K2_50	SA-182 F11	44.000	10.000	...	...	...
K3_80	SA-182 F11	44.000	10.000	...	...	...
K4_80	SA-182 F11	44.000	10.000	...	...	...
N2_600	SA-182 F11	44.000	10.000	...	...	...
UC-1	SA-182 F11	84.000	10.000	...	...	...
M1_600	SA-182 F11	44.000	10.000	...	...	...
N1_600	SA-182 F11	44.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
K1_50	...	240.0	1000.00	0.00	Top head
K2_50	...	120.0	300.00	0.00	Top head
K3_80	...	0.0	500.00	0.00	Bottom head
K4_80	...	180.0	2000.00	0.00	Top head
N2_600	...	0.0	300.00	300.00	Bottom head
UC-1	90.000	0.0	350.00	0.00	Shell
M1_600	...	0.0	1000.00	0.00	Top head
N1_600	...	0.0	1000.00	0.00	Top head

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	SPHERICAL HEAD: 49.0mm THK X 5389.0mm ID X 0.0cm	SA-387 11
1	INSULATION: 2675mm X 100mm THK	
1	PACKING: 2900.000mm	...
1	INSULATION: 2900mm X 100mm THK	
1	INSULATION: 2693mm X 100mm THK	
2	CLASS 300 GR 1.9, 600.0mm BLIND FLANGE(S)	SA-182 F11
2	CLASS 300 GR 1.9, 50.0mm BLIND FLANGE(S)	SA-182 F11
1	CLASS 300 GR 1.9, 80.0mm BLIND FLANGE(S)	SA-182 F11
1	BASERING: 5742.0mm OD X 5270.0 ID X 38.0mm	SA-36
24	BASERING BOLTS: 48.0mm DIA	SA-307 A
24	BASERING BOLT NUTS: 48.0mm DIA	...
1	BASERING TOP RING: 5742.0mm OD X 5462.0 ID X 44.0mm	...
48	BASERING GUSSET PLATES: 140.0mm X 140.0mm X 268.0mm X 16.0mm	...
4	CYLINDRICAL SKIRT SEGMENT 22.0mm THK X 3073.2mm X 320.0cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 22.0mm THK X 3120.5mm X 320.0cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 22.0mm THK X 1607.5mm X 320.0cm	SA-516 70
3	CYLINDRICAL SKIRT SEGMENT 22.0mm THK X 3782.4mm X 350.0cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 22.0mm THK X 3829.7mm X 350.0cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 22.0mm THK X 1843.9mm X 350.0cm	SA-516 70
3	CYLINDRICAL SKIRT SEGMENT 22.0mm THK X 3782.4mm X 120.0cm	SA-387 11
1	CYLINDRICAL SKIRT SEGMENT 22.0mm THK X 3829.7mm X 120.0cm	SA-387 11
1	CYLINDRICAL SKIRT SEGMENT 22.0mm THK X 1843.9mm X 120.0cm	SA-387 11
2	CYLINDRICAL SEGMENT 84.0mm THK X 6676.2mm X 290.0cm	SA-387 11
1	CYLINDRICAL SEGMENT 84.0mm THK X 3454.8mm X 290.0cm	SA-387 11
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Bottom head	[9]	A	33	32		0.980	44.000	1.00	Yes
Shell	[8]	A	46	46		1.000	84.000	1.00	Yes
Top head	[9]	A	33	32		0.980	44.000	1.00	Yes
N2600	[1]	A	33	32		0.974	44.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-26					
K380	[1]	A	-29	-104		0.181	9.735	1.00	Yes
Nozzle Flg	[5]	B	-18	-26					
UC-1	[1]	A	46	45		0.993	84.000	1.00	No
Nozzle Flg	[5]	B	-18	-26					
M1600	[1]	A	33	32		0.974	44.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-26					
K150	[1]	A	-29	-104		0.113	9.690	1.00	Yes
Nozzle Flg	[5]	B	-18	-26					
N1600	[1]	A	33	32		0.974	44.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-26					
K250	[1]	A	-29	-104		0.113	9.690	1.00	Yes
Nozzle Flg	[5]	B	-18	-26					
K480	[1]	A	33	32		0.974	44.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-26					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	No

Warmest MDMT: 46 46

Required Minimum Design Metal Temperature -12.0 °C
 Warmest Computed Minimum Design Metal Temperature 46.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and
 UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
 The Basic MDMT includes the (30F) PWHT credit if applicable.



Class From To : Basic Element Checks.

Note	20	40	There is a high jump in the Joint Eff.
Note	30	50	There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (Kgfc^{m2}):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in. ⁴	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in. ³	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	44.531	kgf/cm ²
Design Internal Temperature	275.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-12.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	Y	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16		IP+VE+EW	
Load Case 17		NP+VO+OW	
Load Case 18		FS+BS+IP+OW	
Load Case 19		FS+BS+EP+OW	
Load Case 20		BL+IP+OW	
Wind Design Code		ASCE/SEI 7-16	
Wind Load Reduction Scale Factor		1.100	
Basic Wind Speed	[V]	97.2	km/hr
Surface Roughness Category		C: Open Terrain	
Type of Surface		Moderately Smooth	
Base Elevation		0	cm
Percent Wind for Hydrotest		50.0	
Using User defined Wind Press. Vs Elev.		N	
Height of Hill or Escarpment	H or Hh	0	cm
Distance Upwind of Crest	Lh	0	cm
Distance from Crest to the Vessel	x	0	cm
Type of Terrain (Hill, Escarpment)		Flat	
Damping Factor (Beta) for Wind (Ope)		0.0100	
Damping Factor (Beta) for Wind (Empty)		0.0000	
Damping Factor (Beta) for Wind (Filled)		0.0000	
Site Elevation above Sea Level		0	cm
Seismic Design Code		ASCE/SEI 7-16	
Seismic Load Reduction Scale Factor		1.000	
Importance Factor		2.000	
Table Value Fa		1.600	
Table Value Fv		2.400	
Max. Mapped Res. Acceleration	[Ss]	0.147	
Max. Eff. Ground Acceleration	[S]	0.065	
Force Modification Factor R		2.000	
Site Class		D	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	2.500	
Force Factor		0.000	
Consider Vertical Acceleration		Yes	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Moment Reduction Factor Tau		1.000	
Design Pressure + Static Head		Y	
Consider MAP New and Cold in Noz. Design		N	
Consider External Loads for Nozzle Des.		Y	
Use ASME VIII-1 Appendix 1-9		N	
Perform Blast Load Analysis		No	
Material Database Year	Current w/Addenda or Code Year		

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/6
Element From Node	10



Element To Node	20	
Element Type	Skirt Support	
Description	Skirt-3	
Distance "FROM" to "TO"	320	cm
Skirt Outside Diameter	5462	mm
Diameter of Skirt at Base	5462	mm
Skirt Thickness	22	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	22	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	275	°C
Design Temperature External Pressure	275	°C
Effective Diameter Multiplier	1.22	
Material Name	SA-516 70	
Allowable Stress, Ambient	1407.1	kgf/cm ²
Allowable Stress, Operating	1396.9	kgf/cm ²
Allowable Stress, Hydrotest	2386	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	2141.3	kgf/cm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Head-to-Skirt or Circ. Seam	1.0	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Weight	
Detail ID	Tailing lug	
Dist. from "FROM" Node / Offset dist	-0.58217E-04	cm
Miscellaneous Weight	1500	kgf
Offset from Element Centerline	2675	mm

Element#	2/6	
Element From Node	20	
Element To Node	30	
Element Type	Skirt Support	
Description	Skirt-2	
Distance "FROM" to "TO"	350	cm
Skirt Outside Diameter	5462	mm
Diameter of Skirt at Base	5462	mm
Skirt Thickness	22	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	22	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	275	°C
Design Temperature External Pressure	275	°C
Effective Diameter Multiplier	1.22	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Head-to-Skirt or Circ. Seam	1.0	
Weld is pre-Heated	No	

Element#	3/6	
Element From Node	30	
Element To Node	40	
Element Type	Skirt Support	
Description	Skirt-1	
Distance "FROM" to "TO"	120	cm
Skirt Outside Diameter	5462	mm
Diameter of Skirt at Base	5462	mm
Skirt Thickness	22	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	22	mm



External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	275	°C
Design Temperature External Pressure	275	°C
Effective Diameter Multiplier	1.22	
Material Name	SA-387 11	
Allowable Stress, Ambient	1509.1	kgf/cm ²
Allowable Stress, Operating	1509.1	kgf/cm ²
Allowable Stress, Hydrotest	2844.9	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	16.002	mm
Yield Stress, Operating	2539	kgf/cm ²
UCS-66 Chart Curve Designation	A	
External Pressure Chart Name	CS-3	
UNS Number	K11789	
Class / Thickness / Grade	2 ::	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.5	
Weld is pre-Heated	No	

Element#	4/6	
Element From Node	40	
Element To Node	50	
Element Type	Spherical	
Description	Bottom head	
Distance "FROM" to "TO"	0	cm
Inside Diameter	5389	mm
Element Thickness	44	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	49	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	44.531	kgf/cm ²
Design Temperature Internal Pressure	275	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	275	°C
Effective Diameter Multiplier	1.22	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 10	
Dist. from "FROM" Node / Offset dist	-267.5	cm
Height/Length of Insulation	267.5	cm
Thickness of Insulation	100	mm
Density	0.000352	kg/cm ³

Element From Node	40	
Detail Type	Nozzle	
Detail ID	N2600	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	600	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-182 F11	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K380	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	80	mm
Nozzle Schedule	160	



Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-182 F11	

Element From Node	40	
Detail Type	Weight	
Detail ID	Closure plate	
Dist. from "FROM" Node / Offset dist	-269.3	cm
Miscellaneous Weight	250	kgf
Offset from Element Centerline	0	mm

Element From Node	40	
Detail Type	Weight	
Detail ID	iNERT BALLS	
Dist. from "FROM" Node / Offset dist	-22.5	cm
Miscellaneous Weight	9999.8	kgf
Offset from Element Centerline	0	mm

Element From Node	40	
Detail Type	Weight	
Detail ID	FLOATING SCREEN	
Dist. from "FROM" Node / Offset dist	0	cm
Miscellaneous Weight	99.998	kgf
Offset from Element Centerline	0	mm

Element#	5/6	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	Shell	
Distance "FROM" to "TO"	290	cm
Inside Diameter	5350	mm
Element Thickness	84	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	84	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	44.531	kgf/cm ²
Design Temperature Internal Pressure	275	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	275	°C
Effective Diameter Multiplier	1.22	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Packing	
Detail ID	Catalyst	
Dist. from "FROM" Node / Offset dist	0	cm
Height of Packed Section	290	cm
Density	0.0006055	kg/cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	

Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	290	cm
Thickness of Insulation	100	mm
Density	0.000352	kg/cm ³

Element From Node	50	
Detail Type	Nozzle	



Detail ID	UC-1	
Dist. from "FROM" Node / Offset dist	90	cm
Nozzle Diameter	600	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-182 F11	

Element From Node	50	
Detail Type	Weight	
Detail ID	SF wt2	
Dist. from "FROM" Node / Offset dist	0	cm
Miscellaneous Weight	1500	kgf
Offset from Element Centerline	0	mm

Element From Node	50	
Detail Type	Weight	
Detail ID	Clip wt	
Dist. from "FROM" Node / Offset dist	145	cm
Miscellaneous Weight	1550	kgf
Offset from Element Centerline	0	mm

Element From Node	50	
Detail Type	Weight	
Detail ID	misc margin	
Dist. from "FROM" Node / Offset dist	0	cm
Miscellaneous Weight	10000	kgf
Offset from Element Centerline	1500	mm

Element#	6/6	
Element From Node	60	
Element To Node	70	
Element Type	Spherical	
Description	Top head	
Distance "FROM" to "TO"	0	cm
Inside Diameter	5389	mm
Element Thickness	44	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	49	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	44.531	kgf/cm ²
Design Temperature Internal Pressure	275	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	275	°C
Effective Diameter Multiplier	1.22	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	269.3	cm
Thickness of Insulation	100	mm
Density	0.000352	kg/cm ³

Element From Node	60	
Detail Type	Nozzle	
Detail ID	M1600	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	600	mm
Nozzle Schedule	160	
Nozzle Class	300	



Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-182 F11	
Element From Node	60	
Detail Type	Nozzle	
Detail ID	K150	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	50	mm
Nozzle Schedule	XXS	
Nozzle Class	300	
Layout Angle	240.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-182 F11	
Element From Node	60	
Detail Type	Nozzle	
Detail ID	N1600	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	600	mm
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-182 F11	
Element From Node	60	
Detail Type	Nozzle	
Detail ID	K250	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	50	mm
Nozzle Schedule	XXS	
Nozzle Class	300	
Layout Angle	120.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-182 F11	
Element From Node	60	
Detail Type	Nozzle	
Detail ID	K480	
Dist. from "FROM" Node / Offset dist	2000	mm
Nozzle Diameter	80	mm
Nozzle Schedule	XXS	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	78.622	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-182 F11	
Element From Node	60	
Detail Type	Weight	
Detail ID	N1Vane distrib	
Dist. from "FROM" Node / Offset dist	267.5	cm
Miscellaneous Weight	299.99	kgf
Offset from Element Centerline	0	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	M1Davitt	
Dist. from "FROM" Node / Offset dist	240	cm
Miscellaneous Weight	99.998	kgf



Offset from Element Centerline	1500	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	SF wt-1	
Dist. from "FROM" Node / Offset dist	0	cm
Miscellaneous Weight	1500	kgf
Offset from Element Centerline	0	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	FLOATING BASKET	
Dist. from "FROM" Node / Offset dist	7.5	cm
Miscellaneous Weight	99.998	kgf
Offset from Element Centerline	0	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	INERT BALLS-TOP	
Dist. from "FROM" Node / Offset dist	22.5	cm
Miscellaneous Weight	4349.9	kgf
Offset from Element Centerline	0	mm
Element From Node	60	
Detail Type	For./Mom.	
Detail ID	N2F/M	
Dist. from "FROM" Node / Offset dist	269.3	cm
Force in X Direction	4399.9	kgf
Force in Y Direction	-6054.9	kgf
Force in Z Direction	4399.9	kgf
Moment about X Axis	8073.7	kgf-m.
Moment about Y Axis	0	kgf-m.
Moment about Z Axis	8073.7	kgf-m.
Force/Moment Combination Method	SRSS	

XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
	Skirt-3	...	320	...	320
	Skirt-2	...	670	...	350
	Skirt-1	...	790	...	120
	Bottom head	...	790	...	...
	Shell	...	1080	...	290
	Top head	...	1080	...	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
Skirt-3	...	...	22	1.6	5462	...	1
Skirt-2	...	...	22	1.6	5462	...	2
Skirt-1	...	...	22	1.6	5462	...	3
Bottom head		44.531	49	3	5389	1509.1	4
Shell		44.531	84	3	5350	1509.1	5
Top head		44.531	49	3	5389	1509.1	6

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
Skirt-3	...	...	No Calc	No Calc	22	No Calc	1
Skirt-2	...	...	No Calc	No Calc	22	No Calc	2
Skirt-1	...	...	No Calc	No Calc	22	No Calc	3
Bottom head		44.531	45.7354	49.1255	44	42.9171	4
Shell		44.531	44.8313	46.5122	84	83.4476	5
Top head		44.531	45.7354	49.1255	44	42.9171	6

Minimum 45 47

MAWP: 45 kgf/cm², limited by: Shell.

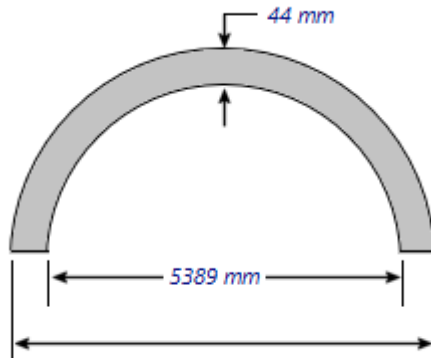
Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Bottom head

Spherical Head Design Information:					
Design Pressure	P	44.531 kgf/cm²	Int. Design Temperature	275 °C	
Material		SA-387 11	Ext. Design Temperature	275 °C	
Length		0.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1509.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm²
Inside Diameter	Di	5389.000 mm	Actual Stress	1469.4 kgf/cm²	
Spec. Min. Thk t _e , or s _a		44.0000 mm	Spec. Nominal Thk.	49.000 mm	
Surface Area		472922.3 cm²	SG of Contents	0.000	
Element Volume		40972572.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		17641.2 kgf	Operating Weight	17641.2 kgf	
			Comp. All. Ope. Stress	931.0 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 67.363 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 33.681 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)}$$

$$= (44.531 \cdot 2697.5) / (2 \cdot 1509.1 \cdot 1.0 - 0.2 \cdot 44.531)$$

$$= 39.9171 + 3.0000 = 42.9171 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 1509.1 \cdot 1.0 \cdot 41.0) / (2697.5 + 0.2 \cdot 41.0)$$

$$= 46 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 1509.1 \cdot 1.0 \cdot 44.0) / (2694.5 + 0.2 \cdot 44.0)$$

$$= 49 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (44.531 \cdot (2697.5 + 0.2 \cdot 41.0)) / (2 \cdot 1.0 \cdot 41.0)$$



= 1469.360 kgf/cm²

Percent Elong. per UCS-79, VIII-1-01-57 (75*tnom/Rf)*(1-Rf/Ro) 1.352 %

MDMT Calculations in the Spherical Portion:

Govrn. thk, tg = 44.0, tr = 40.187, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.98, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 33 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 32 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	44.831 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	44.000 mm	Basic MDMT	33 °C
Stress/Thk. Ratio	0.980	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	32 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 67.363 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 33.681 mm

EXTERNAL PRESSURE RESULTS for: Bottom head

Spherical Head From 40 to 50 Ext. Chart: CS-2 at 275 °C

Elastic Modulus from Chart: CS-2 at 275 °C: 1874597 kgf/cm²

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Sphere OD	Do/t	Factor A	Factor B
41.000	5477.00	133.59	0.0018715	775.84

$$MAEP = B / ((Do/t)/2) = 775.8367 / ((133.5854/2)) = 11.6156 \text{ kgf/cm}^2$$

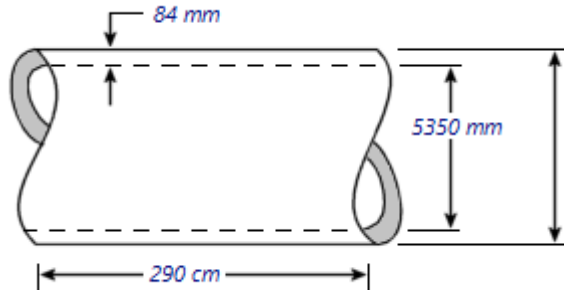
Results for Required Thickness				Tca
Tca	Sphere OD	Do/t	Factor A	Factor B
8.216	5477.00	666.60	0.0003750	351.52

$$MAEP = B / ((Do/t)/2) = 351.5231 / ((666.5982/2)) = 1.0547 \text{ kgf/cm}^2$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell

Cylindrical Shell Design Information:					
Design Pressure	P	44.531 kgf/cm²	Int. Design Temperature	275 °C	
Material		SA-387 11	Ext. Design Temperature	275 °C	
Length		290.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1509.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm²
Inside Diameter	Di	5350.000 mm	Actual Stress	1499.0 kgf/cm²	
Spec. Min. Thk t _e , or s _a		84.0000 mm	Spec. Nominal Thk.	84.000 mm	
Surface Area		502724.0 cm²	SG of Contents	0.000	
Element Volume		65192176.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		32230.6 kgf	Operating Weight	32230.6 kgf	
			Comp. All. Ope. Stress	1049.8 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (44.531 \cdot 2678.0) / (1509.1 \cdot 1.0 - 0.6 \cdot 44.531)$$

$$= 80.4476 + 3.0000 = 83.4476 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1509.1 \cdot 1.0 \cdot 81.0) / (2678.0 + 0.6 \cdot 81.0)$$

$$= 45 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1509.1 \cdot 1.0 \cdot 84.0) / (2675.0 + 0.6 \cdot 84.0)$$

$$= 47 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (44.531 \cdot (2678.0 + 0.6 \cdot 81.0)) / (1.0 \cdot 81.0)$$

$$= 1498.990 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.546 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 84.0, tr = 81.0, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 1.0, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 46 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	44.831 kgf/cm ²
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	84.000 mm	Basic MDMT	46 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	46 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell

Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 275 °C

Elastic Modulus from Chart: CS-2 at 275 °C: 1874597 kgf/cm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
81.000	5518.00	4696.33	68.12	0.8511	0.0029126	841.08

$$MAEP = (4*B)/(3*(Do/t)) = (4*841.0823)/(3*68.1235) = 16.4619 \text{ kgf/cm}^2$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
17.083	5518.00	4696.33	323.02	0.8511	0.0002726	255.50

$$MAEP = (4*B)/(3*(Do/t)) = (4*255.5045)/(3*323.0192) = 1.0547 \text{ kgf/cm}^2$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
81.000	5518.00	132518.78	68.12	24.0157	0.0002460	230.59

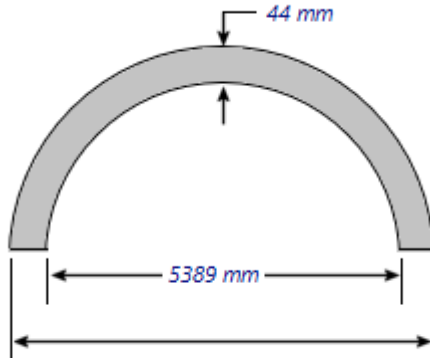
$$MAEP = (4*B)/(3*(Do/t)) = (4*230.5918)/(3*68.1235) = 4.5132 \text{ kgf/cm}^2$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Top head

Spherical Head Design Information:					
Design Pressure	P	44.531 kgf/cm²	Int. Design Temperature	275 °C	
Material		SA-387 11	Ext. Design Temperature	275 °C	
Length		0.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1509.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm²
Inside Diameter	Di	5389.000 mm	Actual Stress	1469.4 kgf/cm²	
Spec. Min. Thk t _e , or s _a		44.0000 mm	Spec. Nominal Thk.	49.000 mm	
Surface Area		472922.3 cm²	SG of Contents	0.000	
Element Volume		40972572.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		17641.2 kgf	Operating Weight	17641.2 kgf	
			Comp. All. Ope. Stress	931.0 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 67.363 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 33.681 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)}$$

$$= (44.531 \cdot 2697.5) / (2 \cdot 1509.1 \cdot 1.0 - 0.2 \cdot 44.531)$$

$$= 39.9171 + 3.0000 = 42.9171 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 1509.1 \cdot 1.0 \cdot 41.0) / (2697.5 + 0.2 \cdot 41.0)$$

$$= 46 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 1509.1 \cdot 1.0 \cdot 44.0) / (2694.5 + 0.2 \cdot 44.0)$$

$$= 49 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (44.531 \cdot (2697.5 + 0.2 \cdot 41.0)) / (2 \cdot 1.0 \cdot 41.0)$$



= 1469.360 kgf/cm²

Percent Elong. per UCS-79, VIII-1-01-57 (75*tnom/Rf)*(1-Rf/Ro) 1.352 %

MDMT Calculations in the Spherical Portion:

Govrn. thk, tg = 44.0, tr = 40.187, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.98, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 33 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 32 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	44.831 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	44.000 mm	Basic MDMT	33 °C
Stress/Thk. Ratio	0.980	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	32 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 67.363 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 33.681 mm

EXTERNAL PRESSURE RESULTS for: Top head

Spherical Head From 60 to 70 Ext. Chart: CS-2 at 275 °C

Elastic Modulus from Chart: CS-2 at 275 °C: 1874597 kgf/cm²

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Sphere OD	Do/t	Factor A	Factor B
41.000	5477.00	133.59	0.0018715	775.84

$$MAEP = B / ((Do/t)/2) = 775.8367 / ((133.5854/2)) = 11.6156 \text{ kgf/cm}^2$$

Results for Required Thickness				Tca
Tca	Sphere OD	Do/t	Factor A	Factor B
8.216	5477.00	666.60	0.0003750	351.52

$$MAEP = B / ((Do/t)/2) = 351.5231 / ((666.5982/2)) = 1.0547 \text{ kgf/cm}^2$$

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	58.281 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	57.890 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	59.927 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	49.314 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	63.679 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	63.679 kgf/cm ²
App 27-4	= M.A.W.P.	44.831 kgf/cm ²

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	1857.5	2844.9	0.653	60.47
Shell	1961.7	2844.9	0.690	60.46
Top head	1857.5	2844.9	0.653	60.47

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2_600	Nozzle	1407.00	1407.00	1.000
K3_80	Nozzle	1407.00	1407.00	1.000
UC-1	Nozzle	1407.00	1407.00	1.000
M1_600	Nozzle	1407.00	1407.00	1.000
K1_50	Nozzle	1407.00	1407.00	1.000
N1_600	Nozzle	1407.00	1407.00	1.000
K2_50	Nozzle	1407.00	1407.00	1.000
K4_80	Nozzle	1407.00	1407.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
Bottom head	1509.10	1509.10	1.000
Shell	1509.10	1509.10	1.000
Top head	1509.10	1509.10	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
N2_600	882.80	2524.00	0.350
K3_80	251.91	2524.00	0.100
UC-1	943.21	2524.00	0.374
M1_600	329.74	2524.00	0.131
K1_50	164.03	2524.00	0.065
N1_600	656.64	2524.00	0.260
K2_50	164.03	2524.00	0.065
K4_80	177.37	2524.00	0.070

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	58.281 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	57.890 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	59.907 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	49.314 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	63.679 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	63.679 kgf/cm ²
App 27-4	= M.A.W.P.	44.831 kgf/cm ²

UG-99(b) Note 35, Test Pressure Calculation:
 = Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 44.531 • 1.0
 = 57.890 kgf/cm²

Vertical Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	1937.5	2844.9	0.681	58.72
Shell	1967.5	2844.9	0.692	58.45
Top head	1919.1	2844.9	0.675	58.16

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2_600	Nozzle	1407.00	1407.00	1.000
K3_80	Nozzle	1407.00	1407.00	1.000
UC-1	Nozzle	1407.00	1407.00	1.000
M1_600	Nozzle	1407.00	1407.00	1.000
K1_50	Nozzle	1407.00	1407.00	1.000
N1_600	Nozzle	1407.00	1407.00	1.000
K2_50	Nozzle	1407.00	1407.00	1.000
K4_80	Nozzle	1407.00	1407.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
Bottom head	1509.10	1509.10	1.000
Shell	1509.10	1509.10	1.000
Top head	1509.10	1509.10	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
N2_600	857.29	2524.00	0.340
K3_80	244.63	2524.00	0.097
UC-1	911.81	2524.00	0.361
M1_600	317.16	2524.00	0.126
K1_50	157.77	2524.00	0.063
N1_600	631.59	2524.00	0.250
K2_50	157.77	2524.00	0.063
K4_80	170.60	2524.00	0.068



Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	790	...	...	No Calc	No Calc
20	30	790	...	...	No Calc	No Calc
30	40	790	...	...	No Calc	No Calc
40	50	No Calc	5477	41	0.0018715	775.837
50	60	469.633	5518	81	0.0029126	841.082
60	70	No Calc	5477	41	0.0018715	775.837

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	...	No Calc	...	No Calc
40	50	44	11.2163	1.05461	11.6156
50	60	84	20.0826	1.05461	16.4619
60	70	44	11.2163	1.05461	11.6156

Minimum

12

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	790	No Calc	No Calc	No Calc
20	30	790	No Calc	No Calc	No Calc
30	40	790	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	469.633	13251.9	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	11590.8	...	10912.6	...	...
20	30	10199.1	...	9457.42	...	...
30	40	3496.83	...	3242.54	...	...
40	50	17641.2	40972572	16579.3	41109576	...
50	60	32230.6	65192176	31096.7	65338484	...
60	70	17641.2	40972572	16579.3	41109576	...
Total		92799	0.147E+09	87867	0.148E+09	0

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset Dtl. Cent. cm	Description
10	Wght	1499.97	...	0.58217E-04	-267.5	Tailing lug
40	Ins1	1871.36	...	-133.75	...	Ins: 10
40	Noz1	822.421	...	-267.25	...	N2_600
40	Noz1	27.0845	150	-246.638	...	K3_80
40	Wght	249.996	...	-269.3	...	Closure plate
40	Wght	9999.82	...	-22.5	...	iNERT BALLS
40	Wght	99.9982	...	...	...	FLOATING SCREEN
50	Pack	39519.4	...	145	...	Catalyst
50	Ins1	1803.74	...	145	...	Ins: 20
50	Noz1	1522.18	285.001	90	...	UC-1
50	Wght	1499.97	...	...	...	SF wt2
50	Wght	1550	...	145	...	Clip wt
50	Wght	10000	150	...	...	misc margin
60	Ins1	1871.36	...	134.65	...	Ins: 50
60	Noz1	1968.19	150	271.638	...	M1_600
60	Noz1	26.5055	-75	271.638	129.904	K1_50
60	Noz1	1106.57	...	317.25	...	N1_600
60	Noz1	18.1432	-75	236.638	-129.904	K2_50
60	Noz1	78.6216	-200	278.364	...	K4_80
60	Wght	299.995	...	267.5	...	N1_Vane distrib
60	Wght	99.9982	150	240	...	M1_Davit
60	Wght	1499.97	...	...	...	SF wt-1
60	Wght	99.9982	...	7.5	...	FLOATING BASKET
60	Wght	4349.92	...	22.5	...	INERT BALLS-TOP
60	Forc	...	...	269.3	...	N2_F/M

Total Weight of Each Detail Type:

Packing	39519.4
Insulation	5546.4
Nozzles	5569.7
Weights	31249.6
Sum of the Detail Weights	81885.2 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	92799.7	92799.7	92799.7	92799.7	92799.7	92799.7
Nozzles	5569.7	5569.7	5569.7	5569.7	5569.7	5569.7
Wld Weights	16899.9	16899.9	16899.9	16899.9	16899.9	16899.9
Packing	...	...	39519.4	39519.4	39519.4	39519.4
Insulation	...	...	5546.4	5546.4	5546.4	5546.4
Ope Weights	...	...	...	...	...	14349.7
Test Liquid	...	147071.2	...	...	...	...



Totals	115269.3	262340.6	160335.1	160335.1	160335.1	174684.9
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Field Installation Options:

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	115269.3 kg
Shop Test Wt.	- Fabricated weight + water (full)	262340.6 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	160335.1 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	160335.1 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	160335.1 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	174684.9 kg
Field Test Wt.	- Empty weight + test medium (full)	267887.0 kg

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	144900.4 kg
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	278983.5 kg
Mass of the Upper 1/3 of the Vertical Vessel (empty)	133956.6 kg

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	549100	544677
20	30	600578	595740
30	40	205913	204254
40	50	472922	456180
50	60	502724	487418
60	70	472922	456180

Total	2804159.5	2744449.0 cm ²
Total	3018.4	2954.1 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg	Total. Ele. Oper. Wgt. kg	Total. Ele. Hydro. Wgt. kg	Total Dtl. Offset Mom. kgf-m.	Oper. Wgt. No Liquid kg
10	20	13090.8	13090.8	13090.8	...	13090.8
20	30	10199.1	10199.1	10199.1	...	10199.1
30	40	3496.83	3496.83	3496.83	...	3496.83
40	50	20712	30711.8	61666.2	40.6268	30711.8
50	60	88125.9	88125.9	113769	19338.2	88125.9
60	70	24710.5	29060.4	65664.7	2911.55	29060.4

Empty Support Force + the Sum of the Y forces	166390.02	kgf
Operating Support Force + the Sum of the Y forces	180739.75	kgf
Hydro Support Force + the Sum of the Y forces	273941.88	kgf

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg	Cumulative Oper. Wgt. kg	Cumulative Hydro. Wgt. kg
10	20	174685	174685	267887
20	30	161594	161594	254796
30	40	151395	151395	244597
40	50	147898	147898	241100
50	60	117186	117186	179434
60	70	29060.4	29060.4	65664.7

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. kgf-m.	Cumulative Oper. Mom. kgf-m.	Cumulative Hydro. Mom. kgf-m.
10	20	22290.4	22290.4	22290.4
20	30	22290.4	22290.4	22290.4
30	40	22290.4	22290.4	22290.4
40	50	22290.4	22290.4	22290.4
50	60	22249.8	22249.8	22249.8
60	70	2911.55	2911.55	2911.55

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Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N2_600	46.55	52.72	275	300	GR 1.9	0.002	46.551	46.551	46.551
K3_80	46.55	52.72	275	300	GR 1.9	...	...	...	...
UC-1	46.55	52.72	275	300	GR 1.9	...	...	...	...
M1_600	46.55	52.72	275	300	GR 1.9	...	...	...	...
K1_50	46.55	52.72	275	300	GR 1.9	...	...	...	...
N1_600	46.55	52.72	275	300	GR 1.9	0.003	46.551	46.551	46.551
K2_50	46.55	52.72	275	300	GR 1.9	...	...	...	...
K4_80	46.55	52.72	275	300	GR 1.9	...	...	...	...
Min Rating	46.551	52.721, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME kgf-m.	Axial Force FE kgf	MAWP PD kgf/cm ²	Maximum Pressure Pmax kgf/cm ²	Pass Fail
N2_600	657.225	0.50	1	0	44.8	46.6	Pass
N1_600	657.225	0.50	2	0	44.8	46.6	Pass

Evaluating Flange on Nozzle [N2600]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1) + 4 \cdot \text{abs}(0) \cdot 657.225 \leq \pi \cdot 657.225^3 \cdot ((46.551 - 44.831) + .500 \cdot 46.551)$$

$$= 0.16929E+02 \leq 0.22293E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2600 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(46.551, 0.500 \cdot 46.551 + 46.551 - 16.9 / (\pi \cdot 657.225^3))$$

$$= \min(46.551, 69.825)$$

$$= 46.551 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [N1600]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(2) + 4 \cdot \text{abs}(0) \cdot 657.225 \leq \pi \cdot 657.225^3 \cdot ((46.551 - 44.831) + .500 \cdot 46.551)$$

$$= 0.26451E+02 \leq 0.22293E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1600 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(46.551, 0.500 \cdot 46.551 + 46.551 - 26.5 / (\pi \cdot 657.225^3))$$

$$= \min(46.551, 69.824)$$

$$= 46.551 \text{ kgf/cm}^2$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 9.54333 Hz.

The Natural Frequency for the Vessel (Ope...) is 9.54333 Hz.

The Natural Frequency for the Vessel (Filled) is 7.78262 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. kgf	X,Z Moment and For Res kgf-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	6222.43	28175

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. kgf	X,Z Moment and For Res kgf-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	6222.43	28175

User Input Forces and Moments:

From Node	Distance From	- - - - Fx	Forces - - - - Fy	- - - - Fz	- - - - Mx	Moments - - - - My	- - - - Mz
60	269.30	4400.	-6055.	4400.	8074.		8074.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	0 cm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 cm
Distance Upwind of Crest Lh	0 cm
Distance from Crest to the Vessel x	0 cm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 cm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.215 \cdot 0.927) / (1 + 1.7 \cdot 3.4 \cdot 0.215)))$
 $= \min(0.85, 0.888)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	9.543 Hz
Natural Frequency of Vessel (Empty)	9.543 Hz
Natural Frequency of Vessel (Test)	7.783 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.526
Structure Height to Diameter ratio	2.570
Height to top of Structure	1363.850 cm

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$; $Zg = 27432.0$ cm

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 160.0 + 0.0 = 160.0$ cm
 $= 5.249$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (5.249 ft.) < 15 ft.
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15 / 27432.0)^{2/9.5}$
 $= 0.849$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[Kzt]$:



$K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$$

$$= 16.0 \text{ psf } [78.119 \text{ kgf/m}^2]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16.0 \cdot 0.85 \cdot 0.526 \cdot 229.526$$

$$= 1642.4 \text{ lbs. } [745.0 \text{ kgf}]$$

Element	Hgt (z) cm	K1	K2	K3	Kz	Kzt	qz kgf/m ²
Skirt-3	160.0	0.000	0.000	0.000	0.849	1.000	78.119
Skirt-2	495.0	0.000	0.000	0.000	0.863	1.000	78.119
Skirt-1	730.0	0.000	0.000	0.000	0.937	1.000	78.119
Bottom head	790.0	0.000	0.000	0.000	0.952	1.000	78.119
Shell	935.0	0.000	0.000	0.000	0.987	1.000	78.119
Top head	1200.5	0.000	0.000	0.000	1.040	1.000	78.119

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, cm
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, cm
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
- L - Total height of structure cm
- Lc - Total length of conical section(s) of vessel, cm
- tb - Uncorroded plate thickness at bottom of vessel, mm
- V30 - Design Wind Speed provided by user, km/hr
- Vc - Critical wind velocity, km/hr
- Vw - Maximum wind speed at top of structure, km/hr
- W - Total corroded weight of structure, kgf
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. kgf
- Z - Maximum amplitude of vibration at top of vessel, mm
- D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, <= 0.00032 (High); 0.00032 < 0.0004 (Probable)
- P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: $0.0/1080.0 = 0.0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[18.15/35.43^2] \cdot 10^4 = 144.528$ [Geometry Violation]

Compute the vibration possibility. If $Vp > 0.0004$ no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 169753 / (1080.0 \cdot 556.368^2)$$

$$= 0.50777E-03$$

Since Vp is > 0.0004 no further vibration analysis is required !

Wind Loads on Masses/Equipment/Piping



ID	Wind Area cm ²	Elevation cm	Pressure kgf/m ²	Force kgf
Tailing lug	0.00	-0.00	78.12	0.00
Closure plate	0.00	520.70	78.12	0.00
iNERT BALLS	0.00	767.50	78.12	0.00
FLOATING SCRE	0.00	790.00	78.12	0.00
SF wt2	0.00	790.00	78.12	0.00
Clip wt	0.00	935.00	78.12	0.00
misc margin	0.00	790.00	78.12	0.00
N1vane distr	0.00	1347.50	78.12	0.00
M1davit	0.00	1320.00	78.12	0.00
SF wt-1	0.00	1080.00	78.12	0.00
FLOATING BASK	0.00	1087.50	78.12	0.00
INERT BALLS-T	0.00	1102.50	78.12	0.00

The Natural Frequency for the Vessel (Ope...) is 9.54333 Hz.

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm ²	Wind Pressure kgf/m ²	Element Wind Load kgf
10	20	160	666.364	213236	78.1189	819.495
20	30	495	666.364	233227	78.1189	896.323
30	40	730	666.364	79963.7	78.1189	307.311
40	50	789.999	692.594	2.11103	78.1189	0.0081129
50	60	935	697.596	202303	78.1189	777.476
60	70	1200.47	692.594	154406	78.1189	593.401

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	2.000
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104\end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
= $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
= $0.020(44.27330.75)$
= 0.343 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
= (1/Natural Frequency) = (1/9.543)
= 0.105

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
= Minimum Value of ($1.7 \cdot 0.343$, 0.105) per 12.8.2
= 0.105

Seismic Response Coefficient per equation 12.8-2 [Cs]:
= $Sds / (R / I)$
= $0.157 / (2.0 / 2.0)$
= 0.157

Maximum value of C_s per equation 12.8-3 [Cs]:
= $Sd1 / (T(R / I_e))$
= $0.104 / (0.105(2.0 / 2.0))$
= 0.993

Minimum value of C_s per equation 15.4-1 [Cs]:
= $\max((0.044 \cdot SDS \cdot I_e), 0.030)$
= $\max((0.044 \cdot 0.1568 \cdot 2.0), 0.030)$
= 0.030

Total Base Shear [V]:
= $C_s \cdot W$ (Equation 12.8-1):
= $0.1568 \cdot 174684.9$
= 27390.586 kgf

Vertical load per 12.4-4, [Ev]:
= $0.2 \cdot Sds \cdot W$
= $0.2 \cdot 0.157 \cdot 174685$



= 5478.1 kgf

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.0

The Natural Frequency for the Vessel (Ope...) is 9.54333 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height cm	Earthquake Weight kgf	Element Ope Load kgf	Element Emp Load kgf	Element Vertical Load kgf
10	20	160	13090.8	388.316	389.641	-576.775
20	30	495	10199.1	935.985	939.18	-987.301
30	40	730	3496.83	473.258	474.874	-1307.15
40	50	789.999	30711.8	4498.13	3043.89	-1416.81
50	60	935	88125.9	15276.2	15328.4	-2379.93
60	70	1080	29060.4	5818.69	4964.61	-5143.56



Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support cm	Cumulative Shr Wind Cas kgf	Cumulative Shr Eqk Cas kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	160	6222.43	6222.43	95377.1	95377.1
20	30	495	6222.43	6222.43	75465.3	75465.3
30	40	730	6222.43	6222.43	53686.8	53686.8
40	50	789.999	6222.43	6222.43	46220	46220
50	60	935	6222.43	6222.43	46219.9	46219.9
60	70	1080	6222.43	6222.43	28175	28175

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support cm	Cumulative Wind Shear kgf	Earthquake Shear kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	160	3394.01	27390.6	22384.4	256929
20	30	495	2574.52	27002.3	12834.8	169900
30	40	730	1678.2	26066.3	5392.52	77030
40	50	789.999	1370.89	25593	3563.11	46035.1
50	60	935	1370.88	21094.9	3563.07	46034.4
60	70	1080	593.401	5818.69	714.877	7009.85

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear kgf	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	6222.43	0.0042734	0.016478	0.99133E-05
20	30	6222.43	0.037815	0.066325	0.18110E-04
30	40	6222.43	0.077531	0.089385	0.20269E-04
40	50	6222.43	0.089385	0.089385	0.20269E-04
50	60	6222.43	0.1192	0.14976	0.21288E-04
60	70	6222.43	0.14976	0.14976	0.21288E-04

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)
 Allowable deflection : 54.000 Actual deflection : 0.150 mm

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear kgf	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	3394.01	0.0010346	0.0038082	0.21805E-05
20	30	2574.52	0.008308	0.013862	0.33945E-05
30	40	1678.2	0.015934	0.018069	0.36022E-05
40	50	1370.89	0.018069	0.018069	0.36022E-05
50	60	1370.88	0.023324	0.028619	0.36599E-05
60	70	593.401	0.028619	0.028619	0.36599E-05

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	1141.6	7134.97
20	30	1141.6	7134.97
30	40	1141.6	7134.97
40	50	1186.53	7415.83
50	60	1195.1	7469.39
60	70	1186.53	7415.83

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 54.000 Actual deflection : 0.029 mm

Total Deflection in the Operating Condition + Applied Forces:
 Allowable deflection : 54.000 Actual deflection : 0.178 mm

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	3759.87	3486.42	509289856	472390368
20	30	3759.87	3486.42	509289856	472390368
30	40	3759.87	3486.42	509289856	472390368
40	50	7510.04	7001.87	1011924224	944484224
50	60	14340	13835.5	1918888320	1853391104
60	70	7510.04	7001.87	1011924224	944484224

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Longitudinal Allowable Stresses:

From	To	Tensile kgf/cm ²	Hydrotest Tensile kgf/cm ²	Compressive kgf/cm ²	Hydrotest Compressive kgf/cm ²
10	20	1676.34	2863.21	-806.019	-1017.72
20	30	1676.34	2863.21	-806.019	-1017.72
30	40	905.46	1706.92	-806.019	-1142.6
40	50	1810.92	3413.83	-931.004	-1389.66
50	60	1810.92	3413.83	-1049.82	-1440.05
60	70	1810.92	3413.83	-931.004	-1389.66

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure kgf/cm ²	Longitudinal Stress External Pressure kgf/cm ²	Longitudinal Stress Hydrotest Pressure kgf/cm ²
10	20	...	...	...
20	30	...	...	...
30	40	...	...	...
40	50	1456	-35.4855	1892.8
50	60	727.23	-18.2284	945.399
60	70	1456	-35.4855	1892.8

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty kgf/cm ²	Wght. Str. Operating kgf/cm ²	Wght. Str. Hydrotest kgf/cm ²	Wght. Str. Emp. Mom. kgf/cm ²	Wght. Str. Opr. Mom. kgf/cm ²
10	20	-50.1044	-50.1044	-76.8373	4.71865	4.71865
20	30	-46.3496	-46.3496	-73.0825	4.71865	4.71865
30	40	-43.4242	-43.4242	-70.1571	4.71865	4.71865
40	50	-21.1227	-21.1227	-21.1227	2.36006	2.36006
50	60	-8.47	-8.47	-8.47	1.20049	1.20049
60	70	-4.15039	-4.15039	-4.15039	0.30827	0.30827

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. kgf/cm ²	Bend. Str. Oper. Wind kgf/cm ²	Bend. Str. Oper. Equ. kgf/cm ²	Bend. Str. Hyd. Wind kgf/cm ²	Bend. Str. Hyd. Equ. kgf/cm ²
10	20	4.71865	4.73855	54.3891	2.36927	...
20	30	4.71865	2.71699	35.966	1.35849	...
30	40	4.71865	1.14154	16.3064	0.57077	...
40	50	2.36006	0.37725	4.8741	0.18863	...
50	60	1.20049	0.19225	2.48379	0.096123	...
60	70	0.30827	0.07569	0.74219	0.037845	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case kgf/cm ²	Vortex Shedding Empty Case kgf/cm ²	Vortex Shedding Test Case kgf/cm ²	Earthquake Empty Case kgf/cm ²
10	20	...	...	...	49.8962
20	30	...	...	...	32.9977
30	40	...	...	...	15.0074
40	50	...	...	...	4.51092
50	60	...	...	...	2.29872
60	70	...	...	...	0.63325

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind kgf/cm ²	Longitudinal Stress Y Forces Seismic kgf/cm ²
------	----	---	--



10	20	-1.73671	-0.16543
20	30	-1.73671	-0.28318
30	40	-1.73671	-0.37493
40	50	-0.86476	-0.20235
50	60	-0.43764	-0.17202
60	70	-0.86476	-0.7346

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded kgf/cm ²	Earthquake For/Mom Corroded kgf/cm ²	Wind For/Mom No Corrosion kgf/cm ²	Earthquake For/Mom No Corrosion kgf/cm ²
10	20	20.1903	20.1903	18.7275	18.7275
20	30	15.9752	15.9752	14.8178	14.8178
30	40	11.3649	11.3649	10.5415	10.5415
40	50	4.89368	4.89368	4.56754	4.56754
50	60	2.4938	2.4938	2.40868	2.40868
60	70	2.98311	2.98311	2.7843	2.7843

Bending Stresses due to Blast Load:

From	To	Bending Stress kgf/cm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...

**Stress Combination Load Cases for Vertical Vessels:****Load Case Definition Key**

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1676.34	-81.49	806.02		0.1011
20		1676.34	-71.50	806.02		0.0887
30		905.46	-62.39	806.02		0.0774
40		1810.92	-29.62	931.00		0.0318
50		1810.92	-12.79	1049.82		0.0122
60		1810.92	-8.38	931.00		0.0090

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	24.54	1676.34	-125.07	806.02	0.0146	0.1552
20	7.06	1676.34	-100.32	806.02	0.0042	0.1245
30		905.46	-74.89	806.02		0.0929
40		1810.92	-33.09	931.00		0.0355
50		1810.92	-14.64	1049.82		0.0139
60		1810.92	-8.81	931.00		0.0095

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1676.34	-81.49	806.02		0.1011
20		1676.34	-71.50	806.02		0.0887



30	905.46	-62.39	806.02	0.0774
40	1810.92	-29.62	931.00	0.0318
50	1810.92	-12.79	1049.82	0.0122
60	1810.92	-8.38	931.00	0.0090

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	29.03	1676.34	-129.57	806.02	0.0173	0.1608
20	10.03	1676.34	-103.29	806.02	0.0060	0.1282
30		905.46	-76.19	806.02		0.0945
40		1810.92	-33.45	931.00		0.0359
50		1810.92	-14.82	1049.82		0.0141
60		1810.92	-8.92	931.00		0.0096

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		2863.21	-83.93	1017.72		0.0825
20		2863.21	-79.16	1017.72		0.0778
30		1706.92	-75.45	1142.60		0.0660
40		3413.83	-23.67	1389.66		0.0170
50		3413.83	-9.77	1440.05		0.0068
60		3413.83	-4.50	1389.66		0.0032

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		2863.21	-81.56	1017.72		0.0801
20		2863.21	-77.80	1017.72		0.0764
30		1706.92	-74.88	1142.60		0.0655
40		3413.83	-23.48	1389.66		0.0169
50		3413.83	-9.67	1440.05		0.0067
60		3413.83	-4.46	1389.66		0.0032

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1676.34	-81.49	806.02		0.1011
20		1676.34	-71.50	806.02		0.0887
30		905.46	-62.39	806.02		0.0774
40	1441.64	1810.92		931.00	0.7961	
50	722.21	1810.92		1049.82	0.3988	
60	1454.35	1810.92		931.00	0.8031	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	29.03	1676.34	-129.57	806.02	0.0173	0.1608
20	10.03	1676.34	-103.29	806.02	0.0060	0.1282
30		905.46	-76.19	806.02		0.0945
40	1446.80	1810.92		931.00	0.7989	
50	724.77	1810.92		1049.82	0.4002	
60	1455.15	1810.92		931.00	0.8035	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1676.34	-81.49	806.02		0.1011
20		1676.34	-71.50	806.02		0.0887
30		905.46	-62.39	806.02		0.0774
40		1810.92	-65.10	931.00		0.0699
50		1810.92	-31.02	1049.82		0.0296
60		1810.92	-43.87	931.00		0.0471



Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	29.03	1676.34	-129.57	806.02	0.0173	0.1608
20	10.03	1676.34	-103.29	806.02	0.0060	0.1282
30		905.46	-76.19	806.02		0.0945
40		1810.92	-68.94	931.00		0.0740
50		1810.92	-33.05	1049.82		0.0315
60		1810.92	-44.40	931.00		0.0477

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		2863.21	-83.93	1017.72		0.0825
20		2863.21	-79.16	1017.72		0.0778
30		1706.92	-75.45	1142.60		0.0660
40	1874.23	3413.83		1389.66	0.5490	
50	938.23	3413.83		1440.05	0.2748	
60	1889.00	3413.83		1389.66	0.5533	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		2863.21	-81.56	1017.72		0.0801
20		2863.21	-77.80	1017.72		0.0764
30		1706.92	-74.88	1142.60		0.0655
40	1874.04	3413.83		1389.66	0.5490	
50	938.13	3413.83		1440.05	0.2748	
60	1888.96	3413.83		1389.66	0.5533	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1676.34	-54.82	806.02		0.0680
20		1676.34	-51.07	806.02		0.0634
30		905.46	-48.14	806.02		0.0597
40	1437.24	1810.92		931.00	0.7937	
50	719.96	1810.92		1049.82	0.3976	
60	1452.16	1810.92		931.00	0.8019	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1676.34	-46.46	806.02		0.0576
20		1676.34	-42.98	806.02		0.0533
30		905.46	-40.27	806.02		0.0500
40	1436.31	1810.92		931.00	0.7931	
50	719.06	1810.92		1049.82	0.3971	
60	1452.13	1810.92		931.00	0.8019	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1676.34	-54.82	806.02		0.0680
20		1676.34	-51.07	806.02		0.0634
30		905.46	-48.14	806.02		0.0597
40	1437.24	1810.92		931.00	0.7937	
50	719.96	1810.92		1049.82	0.3976	
60	1452.16	1810.92		931.00	0.8019	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
-----------	----------------	-------------------	--------------	-------------------	-------------	-------------

10		1676.34	-54.82	806.02		0.0680
20		1676.34	-51.07	806.02		0.0634
30		905.46	-48.14	806.02		0.0597
40	1437.24	1810.92		931.00	0.7937	
50	719.96	1810.92		1049.82	0.3976	
60	1452.16	1810.92		931.00	0.8019	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1676.34	-54.82	806.02		0.0680
20		1676.34	-51.07	806.02		0.0634
30		905.46	-48.14	806.02		0.0597
40		1810.92	-23.48	931.00		0.0252
50		1810.92	-9.67	1049.82		0.0092
60		1810.92	-4.46	931.00		0.0048

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1396.95	-75.18	671.68		0.1119
20		1396.95	-67.33	671.68		0.1002
30		754.55	-59.88	671.68		0.0892
40	1441.93	1509.10		775.84	0.9555	
50	722.28	1509.10		874.85	0.4786	
60	1454.41	1509.10		775.84	0.9638	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1396.95	-75.18	671.68		0.1119
20		1396.95	-67.33	671.68		0.1002
30		754.55	-59.88	671.68		0.0892
40		1509.10	-64.06	775.84		0.0826
50		1509.10	-30.56	874.85		0.0349
60		1509.10	-43.66	775.84		0.0563

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1676.34	-54.82	806.02		0.0680
20		1676.34	-51.07	806.02		0.0634
30		905.46	-48.14	806.02		0.0597
40	1437.24	1810.92		931.00	0.7937	
50	719.96	1810.92		1049.82	0.3976	
60	1452.16	1810.92		931.00	0.8019	

Absolute Maximum of the all of the Stress Ratio's 0.9638

Governing Element: Top head

Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt-3	...	-50.3	4.7	20.2	...	+ Tens
Skirt-3	...	-50.3	-4.7	-20.2	-75.2	- Comp
Skirt-2	...	-46.6	4.7	16.0	...	+ Tens
Skirt-2	...	-46.6	-4.7	-16.0	-67.3	- Comp
Skirt-1	...	-43.8	4.7	11.4	...	+ Tens
Skirt-1	...	-43.8	-4.7	-11.4	-59.9	- Comp



Bottom head	1456.0	-21.3	2.4	4.9	1441.9	+ Tens
Bottom head	1456.0	-21.3	-2.4	-4.9	...	- Comp
Shell	727.2	-8.6	1.2	2.5	722.3	+ Tens
Shell	727.2	-8.6	-1.2	-2.5	...	- Comp
Top head	1456.0	-4.9	0.3	3.0	1454.4	+ Tens
Top head	1456.0	-4.9	-0.3	-3.0	...	- Comp



Skirt Support Analysis Per EN 13445-3 Section 16.12:

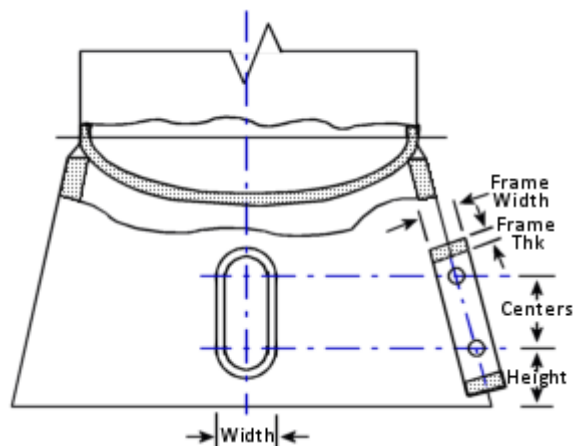
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

The skirt comprises more than one piece. If this is the case, the Openings will only be considered in this Section of the Skirt! Otherwise, make the whole skirt all one piece.

Item: Skirt-2

Total vessel (empty) weight	Fg	149398.125 kgf
Weight of vessel contents (liquid)	Ff	0.000 kgf
Vessel Gross Weight	Fg + Ff	149398.125 kgf
Skirt Mean Diameter	Dz	5440.000 mm
Skirt Analysis Thickness (Corroded)	ez	20.400 mm
Skirt Height	H	3500.000 mm
Additional Axial Force at section 1-1	F1	-6054.894 kgf
Moment at section 1-1	M1	130716.742 kgf-m.
Applied Moment - Bottom of Skirt	Mb	245365.344 kgf-m.
Weight at Bottom of Skirt	W	161594.109 kgf



Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	1.0000	30.0	3000.00	0.00	80.000	16.000
2	1.0000	75.0	700.00	0.00	50.000	8.000
3	1.0000	120.0	3000.00	0.00	80.000	16.000
4	1.0000	175.0	1100.00	0.00	50.000	8.000
5	1.0000	210.0	3000.00	0.00	80.000	16.000
6	1.0000	300.0	3000.00	0.00	80.000	16.000

Interference check:

Access	Access	Pitch mm	Ligament mm
1	2	3139.0366	3138.0366
2	3	3139.0366	3138.0366
3	4	3229.1108	3228.1108
4	5	2524.0178	2523.0178
5	6	4272.4941	4271.4941
1	6	4272.4941	4271.4941



Interference Check: No Overlap Detected between Openings

Minimum Pitch between openings is: 2524.0178 mm
 Minimum Ligament between Openings is: 2523.0178 mm

Recommended minimum Ligament [IMin]:
 $= 2 \cdot \sqrt{(D \cdot t)}$
 $= 2 \cdot \sqrt{(5440.0 \cdot 20.4)} \text{ mm}$
 $= 666.261 \text{ mm}$

Carefully evaluate this acceptance criteria.

Section properties at least metal area [A]:

Number of Openings At Critical Height = 6

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
1	29.9895	30.0105
3	119.9895	120.0105
5	209.9895	210.0105
6	299.9895	300.0105

The points for integrating Ixx, Iyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
30.011	0.000
0.000	119.989
120.011	0.000
0.000	209.989
210.011	299.989
-59.989	29.989

Height of Weakest Section	H	3000.044	mm
Rotation of Principle Axes	a	-0.000	deg.
Centroid Position in X direction	g	12.076	mm
Centroid Position in Y direction	h	-0.000	mm
Principle Moment of Inertia	Ixx	132727408.000	cm ⁴
Principle Moment of Inertia	Iyy	133905896.000	cm ⁴
Distance to Extreme Fiber	ybar	2730.200	mm
Distance to Extreme Fiber	xbar	2742.275	mm
Minimum Section Modulus	W4	486145856.000	mm ³
Metal Area at the Cutting Plane	A4	3603.998	cm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:

$= \sin^{-1}(\text{Largest Opening Width} / D_z)$
 $= \sin^{-1}(1.0/5440.0)$
 $= 0.0002 \text{ rad } (0.0105 \text{ deg})$

Skirt Mean Radius at Section 4-4 [r]:

$= D_z / 2$
 $= 5440.0/2$
 $= 2720.0 \text{ mm}$

Because the Code uses the words 'may be' in the evaluation below, the results should be considered optional.

Check areas for openings per EN 13445 Equation 16.12-76 [A,check]:

At,i	Adelta,i	Status	lt,i <= 8*eat,i
25.6	-3.1	...	...
8.0	-1.4	...	...
25.6	-3.1	...	...
8.0	-1.4	...	...
25.6	-3.1	...	...



25.6 -3.1 ...

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:

$$\begin{aligned} &= \delta \cdot \sqrt{(r / e_z)} \\ &= 0. \cdot \sqrt{(2720.0/20.4)} \\ &= 0.002 \end{aligned}$$

Ok as ratio is < 2 - See EN 13445 16.12.76(a).

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:

$$\begin{aligned} &= e_x(\text{new}) - c_i - c_{\text{ext}} \\ &= 22.0 - 0.8 - 0.8 \\ &= 20.4 \text{ mm} \end{aligned}$$

Find the moment acting at the Critical Height [M4]:

$$\begin{aligned} &= (M_{\text{bottom}} - M_{\text{top}}) \cdot (H - \text{CutHeight}) / H + M_{\text{top}} \\ &= (245365.344 - 130716.742) \cdot (3500.0 - 3000.044) / 3500.0 + 130716.742 \\ &= 147093.641 \text{ kgf-m.} \end{aligned}$$

Compute the section modulus [Zxx and Zyy]:

$$\begin{aligned} Z_{xx} &= I_{xx} / y_{\text{bar}} = 0.13273\text{E}+09 / 2730.2 = 0.48615\text{E}+09 \text{ mm}^3 \\ Z_{yy} &= I_{yy} / x_{\text{bar}} = 0.13391\text{E}+09 / 2742.275 = 0.48830\text{E}+09 \text{ mm}^3 \\ Z &= \min(Z_{xx}, Z_{yy}) = \min(0.48615\text{E}+09, 0.48830\text{E}+09) = 0.48615\text{E}+09 \text{ mm}^3 \end{aligned}$$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:

$$\begin{aligned} &= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area}(A_4))} \\ &= \sqrt{(\min(0.13273\text{E}+09, 0.13391\text{E}+09) / 3603.998)} \\ &= 1919.059 \text{ mm} \end{aligned}$$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:

$$\begin{aligned} &= r_g / L \\ &= 1919.059 / 3500.0 \\ &= 0.548 \end{aligned}$$

Location of the Skirt Centroid at critical height [e]:

$$\begin{aligned} &= \sqrt{(e_x^2 + e_y^2)} \\ &= \sqrt{(12.076^2 + -0.0^2)} \\ &= 12.076 \text{ mm} \end{aligned}$$

$$\text{Where } e_x = 12.076 : e_y = -0.0$$

Moment Increment at Centroid [ΔM_4]:

$$\begin{aligned} &= \text{abs}(F_1 - F_f - F_g) \cdot e = F_4 \cdot e \\ &= \text{abs}(-6054.89 - 149398.12 - 0.0) \cdot 12.076 \\ &= 1877.187 \text{ kgf-m.} \end{aligned}$$

Check of the Skirt in Regions with Openings, per 16.12.4.3:

Weakening Factor, 16.12-70(b) [psi1]:

$$\begin{aligned} &= \min(1; A_4 / (\pi \cdot D_3 \cdot e_{a3})) \\ &= \min(1; 3603.998 / (\pi \cdot 5440.0 \cdot 20.4)) \\ &= 1.000 \end{aligned}$$

Weakening Factor, 16.12-70(c) [psi2]:

$$\begin{aligned} &= \min(1; 4 \cdot W_4 / (\pi \cdot D^3 / (\pi \cdot D_3 \cdot e_{a3}))) \\ &= \min(1; 4.0 \cdot 0.48830\text{E}+09 / (\pi \cdot 5440.0^2 \cdot 20.4)) \\ &= 1.000 \end{aligned}$$

Maximum Bending Moment [Mmax]:

$$\begin{aligned} &= \pi / 4 \cdot d_3^2 \cdot e_{a3} \cdot \sigma_{\text{call}} \\ &= \pi / 4 \cdot 5440.0^2 \cdot 20.4 \cdot 806.0 \\ &= 3821754.5 \text{ kgf-m.} \end{aligned}$$



Maximum Compressive Force [$F_{c,max}$]:

$$\begin{aligned} &= \pi \cdot d_3 \cdot e_{a3} \cdot \sigma_{call} \\ &= \pi \cdot 5440.0 \cdot 20.4 \cdot 806.0 \\ &= 2810110.5 \text{ kgf} \end{aligned}$$

Unity Check of Weakened Section, 16.12-70 [UC1]:

$$\begin{aligned} &= F_4 / (\psi_{i1} \cdot F_{c,max}) + (\text{abs}(M_4) + F_4 \cdot \text{abs}(y_q)) / (\psi_{i2} \cdot M_{max}) \\ &= 155453.0 / (1.0 \cdot 2810111) + (\text{abs}(147093.6) + 155453.0 \cdot \text{abs}(12.076)) / \\ &\quad (1.0 \cdot 3821755) \\ &= 0.094 \text{ must be } \leq \text{ to } 1, \text{ Passed} \end{aligned}$$

The stresses are satisfactory in the region with the opening(s).

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

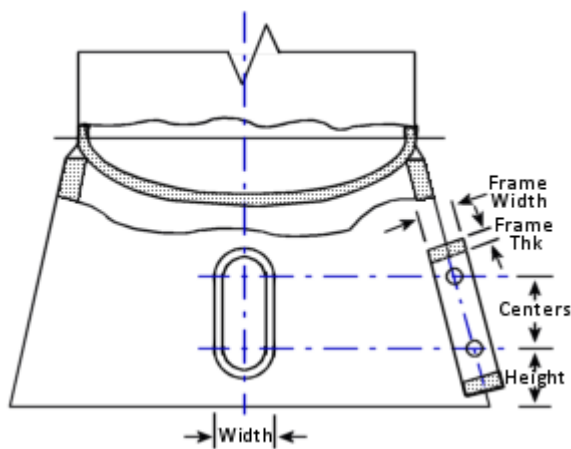
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Skirt-1

Total vessel (empty) weight	Fg	149398.125 kgf
Head Weight below section 2-2	DeltaFg	30711.842 kgf
Weight of vessel contents (liquid)	Ff	0.000 kgf
Vessel Gross Weight	Fg + Ff	149398.125 kgf
Skirt Mean Diameter	Dz	5440.000 mm
Skirt Analysis Thickness (Corroded)	ez	20.400 mm
Skirt Height	H	1200.000 mm
Additional Axial Force at section 1-1	F1	-6054.894 kgf
Moment at section 1-1	M1	92255.156 kgf-m.
Applied Moment - Bottom of Skirt	Mb	130716.742 kgf-m.
Weight at Bottom of Skirt	W	151394.984 kgf

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Spherical
Head Mean Diameter at Skirt Top	Db	5436.000 mm
Head Corroded Thickness	eb	41.000 mm



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	128969272.000 cm ⁴
Moment of Inertia	Iyy =	128969272.000 cm ⁴
Section Modulus	W =	474152256.000 mm ³
Section Metal Area	A =	3486.414 cm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{\frac{\min(I_{xx}, I_{yy})}{\text{Area}}}$$

$$= \sqrt{\frac{\min(0.12897\text{E}+09, 0.12897\text{E}+09)}{3486.414}}$$

$$= 1923.33 \text{ mm}$$

Skirt Slenderness Ratio [rg/L]:

$$= \frac{rg}{L}$$

$$= \frac{1923.33}{1200.0}$$

$$= 1.603$$

Correct the Moment [M4]:

top of skirt is the critical end

Controlling Moment [M4] = 92255.156 kgf-m.



Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$\begin{aligned} &= -M / Z - W / A \\ &= -92255.156 / 18966E+10 - 151394.984 / 3486.41406 \\ &= -48.289 \text{ kgf/cm}^2 \end{aligned}$$

Upwind Side:

$$\begin{aligned} &= +(M) / Z - W / A \\ &= +92255.156 / 18966E+10 - 151394.984 / 3486.41406 \\ &= -38.56 \text{ kgf/cm}^2 \end{aligned}$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$\begin{aligned} &= F1 - Fg - Ff + 4 \cdot M1 / Dz \\ &= -6054.894 - 149398.125 - 0.0 + 4 \cdot 92255.156 / 5440.0 \\ &= -87618.4 \text{ kgf} \end{aligned}$$

Force at the joint, 16.12-2 [Fzq]:

$$\begin{aligned} &= F1 - Fg - Ff - 4 \cdot M1 / Dz \\ &= -6054.894 - 149398.125 - 0.0 - 4 \cdot 92255.156 / 5440.0 \\ &= -223287.6 \text{ kgf} \end{aligned}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$\begin{aligned} &= (Fzp + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb) \\ &= (-87618.414 + 30711.842 + 0.0) / (\pi \cdot 5436.0 \cdot 41.0) + 44.531 \cdot 5436.0 / (4 \cdot 41.0) \\ &= 1467.91 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{1qm}]:

$$\begin{aligned} &= (Fzq + \Delta Fg + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb) \\ &= (-0.2E+06 + 30711.842 + 0.0) / (\pi \cdot 5436.0 \cdot 41.0) + 44.531 \cdot 5436.0 / (4 \cdot 41.0) \\ &= 1448.54 \text{ kgf/cm}^2 \end{aligned}$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$\begin{aligned} &= (Ff + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb) \\ &= (0.0 + 30711.842) / (\pi \cdot 5436.0 \cdot 41.0) + 44.531 \cdot 5436.0 / (4 \cdot 41.0) \\ &= 1480.43 \text{ kgf/cm}^2 \end{aligned}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$\begin{aligned} &= Fzp / (\pi \cdot Dz \cdot ez) \\ &= -87618.414 / (\pi \cdot 5440.0 \cdot 20.4) \\ &= -25.13 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{3qm}]:

$$\begin{aligned} &= Fzq / (\pi \cdot Dz \cdot ez) \\ &= -223287.609 / (\pi \cdot 5440.0 \cdot 20.4) \\ &= -64.05 \text{ kgf/cm}^2 \end{aligned}$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:

$$\begin{aligned} &= 1 - 0.5(Db + eb - Dz + ez) / (r + eb) \\ &= 1 - 0.5(5436.0 + 41.0 - 5440.0 + 20.4) / (2697.5 + 41.0) \\ &= 0.9895 \end{aligned}$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.99) = 8.302 \text{ deg}$$

Dimension [a]:

$$\begin{aligned} &= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))} \\ &= 0.5 \cdot \sqrt{(41.0^2 + 20.4^2 + 2 \cdot 41.0 \cdot 20.4 \cdot 0.99)} \\ &= 30.629 \text{ mm} \end{aligned}$$



Bending moment at verification points p and q [Mp & Mq]:

$$\begin{aligned} M_p &= a \cdot F_{zp} = 30.62 \cdot -87618.4 = -2683.63 \text{ kgf-m.} \\ M_q &= a \cdot F_{zq} = 30.62 \cdot -223287.6 = -6838.98 \text{ kgf-m.} \end{aligned}$$

Correction factor [C]:

$$\begin{aligned} &= 0.63 - 0.057 \left(\frac{e_b}{e_z} \right)^2 \\ &= 0.63 - 0.057 \left(\frac{41.0}{20.4} \right)^2 \\ &= 0.400 \end{aligned}$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot e_b^2) \\ &= 0.4 \cdot 6 \cdot -2683.627 / (\pi \cdot 5436.0 \cdot 41.0^2) \\ &= -22.42 \text{ kgf/cm}^2 \end{aligned}$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot e_b^2) \\ &= 0.4 \cdot 6 \cdot -6838.98 / (\pi \cdot 5436.0 \cdot 41.0^2) \\ &= -57.14 \text{ kgf/cm}^2 \end{aligned}$$

Stress [$\sigma_{3pb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot e_z^2) \\ &= 0.4 \cdot 6 \cdot -2683.627 / (\pi \cdot 5440.0 \cdot 20.4^2) \\ &= -90.50 \text{ kgf/cm}^2 \end{aligned}$$

Stress [$\sigma_{3qb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot e_z^2) \\ &= 0.4 \cdot 6 \cdot -6838.98 / (\pi \cdot 5440.0 \cdot 20.4^2) \\ &= -230.64 \text{ kgf/cm}^2 \end{aligned}$$

Geometry variables, 16.12-32 to 36:

$$\begin{aligned} e_b/D_b &= 1.614/214.016 = 0.0075 \\ y &= 125 \cdot e_b/D_b = 125 \cdot 41.0/5436.0 = 0.9428 \\ \alpha &= 0 \end{aligned}$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= 0.0 \text{ kgf/cm}^2 \text{ (16.12-31)}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$\begin{aligned} &= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)} \\ &= 1467.912 - -22.422 + 0.0 \\ &= 1490.33 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{1potot}]:

$$\begin{aligned} &= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)} \\ &= 1467.912 + -22.422 - 0.0 \\ &= 1445.49 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{1qitot}]:

$$\begin{aligned} &= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)} \\ &= 1448.536 - -57.141 + 0.0 \\ &= 1505.68 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{1qotot}]:

$$\begin{aligned} &= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)} \\ &= 1448.536 + -57.141 - 0.0 \\ &= 1391.40 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{2pitot}]:

$$\begin{aligned} &= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)} \\ &= 1480.426 + -22.422 + 0.0 \\ &= 1458.00 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{2potot}]:



$$\begin{aligned} &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\ &= 1480.426 - -22.422 - 0.0 \\ &= 1502.85 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{2qitot}] &= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)} \\ &= 1480.426 + -57.141 + 0.0 \\ &= 1423.29 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{2qotot}] &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\ &= 1480.426 - -57.141 - 0.0 \\ &= 1537.57 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3pitot}] &= \sigma_{3pm} - \sigma_{3pb} \\ &= -25.131 - -90.503 \\ &= 65.37 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3potot}] &= \sigma_{3pm} + \sigma_{3pb} \\ &= -25.131 + -90.503 \\ &= -115.63 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3qitot}] &= \sigma_{3qm} - \sigma_{3qb} \\ &= -64.045 - -230.639 \\ &= 166.59 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3qotot}] &= \sigma_{3qm} + \sigma_{3qb} \\ &= -64.045 + -230.639 \\ &= -294.68 \text{ kgf/cm}^2 \end{aligned}$$

Allowable stresses:

$$\begin{aligned} \text{For } \sigma_{1pitot} \text{ and } \sigma_{1potot}, \text{ the stress must not exceed:} \\ &= fs(3 - (1/1.5)) \cdot (\sigma_{1pm} / f)^2 \\ &= 1509.1(3 - (1/1.5)) \cdot (1467.912/1509.1)^2 \\ &= 3575.40 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{For } \sigma_{1qitot} \text{ and } \sigma_{1qotot}, \text{ the stress must not exceed:} \\ &= fs(3 - (1/1.5)) \cdot (\sigma_{1qm} / f)^2 \\ &= 1509.1(3 - (1/1.5)) \cdot (1448.536/1509.1)^2 \\ &= 3600.37 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{For } \sigma_{2pitot} \text{ and } \sigma_{2potot}, \text{ the stress must not exceed:} \\ &= fs(3 - (1/1.5)) \cdot (\sigma_{2pm} / f)^2 \\ &= 1509.1(3 - (1/1.5)) \cdot (1480.426/1509.1)^2 \\ &= 3559.10 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{For } \sigma_{2qitot} \text{ and } \sigma_{2qotot}, \text{ the stress must not exceed:} \\ &= fs(3 - (1/1.5)) \cdot (\sigma_{2qm} / f)^2 \\ &= 1509.1(3 - (1/1.5)) \cdot (1480.426/1509.1)^2 \\ &= 3559.10 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{For } \sigma_{3pitot} \text{ \& } \sigma_{3potot}, \text{ the stress must not exceed:} \\ &= fs(3 - (1/1.5)) \cdot (\sigma_{3pm} / f)^2 \\ &= 1509.1(3 - (1/1.5)) \cdot (-25.131/1509.1)^2 \\ &= 4527.02 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{For } \sigma_{3qitot} \text{ \& } \sigma_{3qotot}, \text{ the stress must not exceed:} \\ &= fs(3 - (1/1.5)) \cdot (\sigma_{3qm} / f)^2 \\ &= 1509.1(3 - (1/1.5)) \cdot (-64.045/1509.1)^2 \\ &= 4525.49 \text{ kgf/cm}^2 \end{aligned}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (kgf/cm²)



Stress	Actual	Allowed	Result	Equation
Stot1pi	1490.33	3575.40	Pass	46
Stot1po	1445.49	3575.40	Pass	47
Stot1qi	1505.68	3600.37	Pass	48
Stot1qo	1391.40	3600.37	Pass	49
Stot2pi	1458.00	3559.10	Pass	50
Stot2po	1502.85	3559.10	Pass	51
Stot2qi	1423.29	3559.10	Pass	52
Stot2qo	1537.57	3559.10	Pass	53
Stot3pi	65.37	4527.02	Pass	54
Stot3po	-115.63	4527.02	Pass	55
Stot3qi	166.59	4525.49	Pass	56
Stot3qo	-294.68	4525.49	Pass	57

The stresses are satisfactory in the region without openings.

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**Shop/Field Installation Options :**

Packing is installed in the Shop.

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Packing	935.000 cm
Center of Gravity of the Insulation	935.302 cm
Center of Gravity of the Nozzles	1105.458 cm
Center of Gravity of the Added Weights (Operating)	815.336 cm
Center of Gravity of the Added Weights (Empty)	769.729 cm
Center of Gravity of Bare Shell New and Cold	782.118 cm
Center of Gravity of Bare Shell Corroded	783.826 cm
Vessel CG in the Operating Condition	840.323 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	835.026 cm
Vessel CG in the Test Condition	882.856 cm

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	5462.0000	mm
Skirt Thickness	STHK	22.0000	mm
Skirt Internal Corrosion Allowance	SCA	0.8000	mm
Skirt External Corrosion Allowance		0.8000	mm
Skirt Material		SA-516 70	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	38.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		SA-36	
Basing Ope. All. Stress	BASOPE, f4	1162.42	kgf/cm²
Basing Yield Stress		2528.76	kgf/cm²
Inside Diameter of Basing	DI, D4	5270.0000	mm
Outside Diameter of Basing	DOU, D5	5742.0000	mm

Nominal Diameter of Bolts	BND, d(b0)	48.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Bolt Material		SA-307 A	
Bolt Material Class/Thickness Range		6 <= t <= 100	
Bolt Operating Allowable Stress	SA, fB	1407.00	kgf/cm²
Number of Bolts	RN, nb	24	
Diameter of Bolt Circle	DC, DBC	5602.0000	mm
Bolt Allowable Shear Stress		844.296	kgf/cm²

Ultimate Comp. Strength of Concrete	FPC	418.1	kgf/cm²
Allowable Comp. Strength of Concrete	FC	229.4	kgf/cm²
Modular ratio Steel/Concrete		9.833	

Thickness of Gusset Plates	TGA, e7	16.0000	mm
Width of Gussets at Top Plate	TWDT	140.0000	mm
Width of Gussets at Base Plate	BWDT	140.0000	mm
Gusset Plate Elastic Modulus	E	2057160.0	kgf/cm²
Gusset Plate Yield Stress	SY	2528.8	kgf/cm²
Height of Chair	h1	350.0000	mm
Distance between Gussets	RG, b6	96.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	48.0000	mm
Number of Gussets per bolt	NG	2	

Thickness of Top Plate or Ring	TTA, e5	44.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	140.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	64.0000	mm

External Corrosion Allowance	CA	3.0000	mm
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Dead Weight of Vessel	DW	166390.0	kgf
Operating Weight of Vessel	ROW	180739.8	kgf
Earthquake Moment on Basing	EQMOM	272078.7	kgf-m.
Wind Moment on Basing	WIMOM	37534.5	kgf-m.

User Moment at the Base [Earthquake Case]		95377.1	kgf-m. (UEQMOM)
User Moment at the Base [WindLoad Case]		95377.1	kgf-m. (UWIMOM)
Percent Bolt Preload	ppl	100.0	

Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	

Factor for Increase of Allowables	Fact	1.0000	
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Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$



$$= ((4 \cdot 367456/5602.0) - 175262) / 24$$

$$= 3629.7 \text{ kgf}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 132912/5602.0) - 166390) / 24$$

$$= -2978.6 \text{ kgf}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/5602.0) - 180740) / 24$$

$$= -7530.8 \text{ kgf}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded)	Abss :	11.5517	cm ²
Area Available in all the Bolts	Abss * RN :	277.2397	cm ²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.800	0.701	2.662	1.217	0.316	0.775	13094.2	188355.8	1573.4
4	0.651	0.676	2.335	1.638	0.356	0.783	4051.9	179313.6	361.6
6	0.663	0.657	2.362	1.606	0.353	0.783	4762.0	180023.6	433.4
8	0.654	0.652	2.342	1.631	0.355	0.783	4228.9	179490.5	379.2
9	0.652	0.652	2.338	1.635	0.356	0.783	4140.4	179402.0	370.4

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 4140.4 / (1.575 \cdot 5602.0 \cdot 1.635)$$

$$= 57.408 \text{ kgf/cm}^2, \text{ Should be } \leq 1407.0 \text{ kgf/cm}^2$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 24 \cdot 11.552 / (\pi \cdot 5602.0)$$

$$= 1.575 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 10.961[(2 \cdot 0.6523 \cdot 5602.0 + 236.0) / (2 \cdot 0.6523 \cdot 5602.0)]$$

$$= 11.315 \text{ kgf/cm}^2, \text{ Should be } \leq 229.4 \text{ kgf/cm}^2$$

Values for table 10.3, l = 140.0, b = 366.65, l/b = 0.381835

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(213.82, 878.832) = 878.832 \text{ kgf}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 878.832 / 1669.0)} + 3.0$$

$$= 20.775 \text{ mm}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 140.0, b = 96.0, b/l = 0.685714

As b/l (0.686) is less than 1, inverting b/l = 1.458.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l)) / (\pi \cdot g))) + 1] - [(0.7 - g2)P / (4\pi)]$$

$$= 16253.15 / (4 \cdot \pi) [1.3(\ln((2 \cdot 140.0 \cdot \sin(\pi \cdot 70.0/140.0)) / (\pi \cdot 40.0))) + 1] - [(0.7 - 0.076) \cdot 16253.15 / (4 \cdot \pi)]$$

$$= 1832.7743 \text{ kgf}$$

Required Thickness of Continuous Top Ring [Tc]:



$$= \sqrt{(6 \cdot \text{Abs}(\text{Mo}) / \text{Fb}) + \text{Ca}}$$

$$= \sqrt{(6 \cdot \text{Abs}(1832.77)/1685.84) + 3.0}$$

$$= 28.5401 \text{ mm}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot \text{lavg}) + \text{Ca}$$

$$= 16253.15 / (1265.526 \cdot 140.0) + 3.0$$

$$= 12.525 \text{ (not less than } 9.525 + 3.0 \text{) mm}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 5602.0 / 24 = 733.3 \text{ mm}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76 (P \cdot a / (m \cdot (h + tba) \cdot 1.5 \cdot \text{Sktope}))^{2/3} \cdot r^{1/3} + \text{Ca}$$

$$= 1.76 (16253 \cdot 70.0 / (733.3 \cdot 306.0 \cdot 2095))^{2/3} \cdot 2731.01^{1/3} + \text{Ca}$$

$$= 9.553 + 1.6 = 11.153 \text{ mm}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 33613.0 / (11.552 \cdot 24)$$

$$= 121.2 \text{ kgf/cm}^2. \text{ Must be less than } 844.3 \text{ kgf/cm}^2.$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	20.7747	mm
Actual Basing Thickness as entered by user	38.0000	mm
Required Thickness of Chair Cap	28.5401	mm
Actual Top Ring Thickness as entered by user	44.0000	mm
Required Gusset thickness, + CA	12.5250	mm
Actual Gusset Thickness as entered by user	16.0000	mm
Required Thickness of Skirt for Local Stress	11.1527	mm
Given Thickness of Skirt	22.0000	mm
Required Gusset Height to meet local stress	71.0895	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA})$$

$$= 129.568 \cdot (22.0 - 1.6)$$

$$= 26.43 \text{ kgf/mm}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 26 / [(0.4 \cdot 2141) \cdot 2 \cdot 0.707]$$

$$= 2.182 \text{ mm}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 16253.2 / (172.0 + 2 \cdot (268.0 + 44.0))$$

$$= 20.419 \text{ kgf/mm}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 16253.2 \cdot 70.0 / (172.0 \cdot (312.0) + 0.6667 \cdot (312.0)^2)$$

$$= 9.596 \text{ kgf/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{20.42^2 + 9.6^2}$$

$$= 22.561 \text{ kgf/mm}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 22.56 / [(0.4 \cdot 2141) \cdot 2 \cdot 0.707]$$



= 1.863 mm

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

= Bolt Load / (2 • TopWth)

= 16253.2 / (2 • 140.0)

= 58.047 kgf/mm

Weld Load [Wh]:

= Bolt Load • e / (2 • Hgt • TopWth)

= 16253.2 • 70.0 / (2 • 312.0 • 140.0)

= 13.023 kgf/mm

Resultant Weld Load [Wr]:

= $\sqrt{Wv^2 + Wh^2}$

= $\sqrt{58.05^2 + 13.02^2}$

= 59.49 kgf/mm

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

= Wr / [(0.4 • Yield) • 2 • 0.707]

= 59.49 / [(0.4 • 2141) • 2 • 0.707]

= 4.912 mm

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	6.3500	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	6.3500	mm
Required Gusset to Top Plate Double Fillet Weld Size	4.9119	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N2600

From: 40

Pressure for Reinforcement Calculations	P	44.531	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	275	°C

Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²

Inside Diameter of Hemispherical Head	D	5389.00	mm
Head Finished (Minimum) Thickness	t	44.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		600.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm

Flange Material		SA-182 F11	
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Hub Height of Integral Nozzle	h	60.0000	mm
Height of Beveled Transition	L'	55.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		75.0000	mm

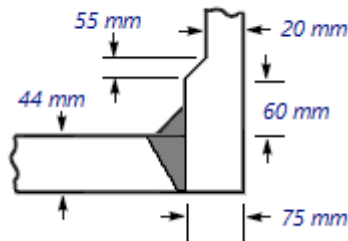
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	44.0000	mm
Inside Projection	h	300.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.9	

Piping Exit Angle [North Clockwise reference]		45.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		4000.0	mm
Centerline Distance to Tangent		4000.0	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N2600

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	600.000 mm.
Actual Thickness Used in Calculation	20.000 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 44.53 \cdot 2697.5 / (2 \cdot 1509 \cdot 1.0 - 0.2 \cdot 44.53)$
 $= 39.9171 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 44.53 \cdot 300.0 / (1407 \cdot 1.0 + 0.4 \cdot 44.53)$
 $= 9.3762 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.4969 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 55.0 / 0.5773$
 $= 95.2629 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1132.0001 mm
Parallel to Vessel Wall, opening length	d	566.0001 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	102.5000 mm
Normal to Vessel Wall, Inward		35.0000 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 1407.0 / 1509.1)$
 $= 0.932$

Weld Strength Reduction Factor [fr2]:

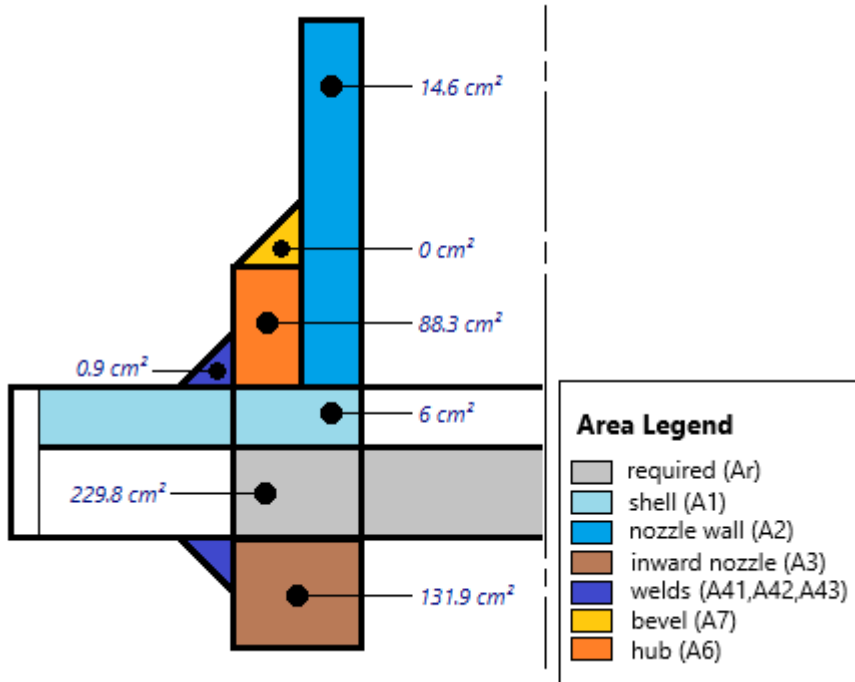
$= \min(1, S_n / S_v)$
 $= \min(1, 1407.0 / 1509.1)$
 $= 0.932$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(0.932, 1.0)$$

$$= 0.932$$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	229.820	23.652	NA
Area in Shell	A1	6.024	182.362	NA
Area in Nozzle Wall	A2	14.571	29.631	NA
Area in Inward Nozzle	A3	131.880	131.880	NA
Area in Welds	A41+A42+A43	0.932	0.932	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	88.281	88.281	NA
TOTAL AREA AVAILABLE	Atot	241.689	433.086	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (566.0 \cdot 39.9171 \cdot 1.0 + 2 \cdot 72.0 \cdot 39.9171 \cdot 1.0 \cdot (1 - 0.93))$$

$$= 229.820 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 566.0 (1.0 \cdot 41.0 - 1.0 \cdot 39.9171) - 2 \cdot 72.0$$

$$(1.0 \cdot 41.0 - 1.0 \cdot 39.9171) \cdot (1 - 0.932)$$

$$= 6.024 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 102.5) (17.0 - 9.38) 0.9323$$

$$= 14.571 \text{ cm}^2$$



Area Available in Inward Nozzle [A3]:

$$= 2 \cdot t_i \cdot \min(h, Tl, 2.5 \cdot t_i) \cdot fr2$$

$$= 2 \cdot 69.0 \cdot (102.5) \cdot 0.9323$$

$$= 131.880 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi\text{-}can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9323 + (0.0)^2 \cdot 0.9323$$

$$= 0.932 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(Tl_{np}, h_o, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(102.5, 300.0, 60.0)) \cdot (75.0 - 20.0) \cdot 0.932$$

$$= 88.281 \text{ cm}^2 \text{ Includes Bevel Area of } 26.747 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 12.3762 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 42.9171 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 42.9171 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.9426 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(42.9171, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(12.3762, 11.3312)$$

$$= 12.3762 \text{ mm}$$

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 361.7,	Allowable	: 1407.0 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3155.8 kgf/cm ²	Passed
Occasional	: 359.8,	Allowable	: 1871.3 kgf/cm ²	Passed
Shear	: 3.8,	Allowable	: 984.9 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 20.0, tr = 9.376, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.552, Temp. Reduction = 27 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-34 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 44.0, tr = 39.917, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.974, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	33 °C
Min Metal Temp. at Required thickness (UCS 66.1)	32 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C



Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -26 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 44.53/52.72 = 0.845

Weld Size Calculations, Description: N2600

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (229.8198 - 6.0236 + 2 • 72.0 • 0.9323 •
(1.0 • 41.0 - 39.9171))1509)
= 339924.50 kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (102.8527 + 0.9323 - 0.0 • 0.93) • 1509
= 156621.86 kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (102.8527 + 131.88 + 0.9323 + (55.0456)) • 1509
= 438710.91 kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (102.8527 + 131.88 + 0.9323 + 0.0 + (55.0456)) • 1509
= 438710.91 kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 600.0 • 10.0 • 0.49 • 1407
= 64977. kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 291.5) • (75.0 - 3.0) • 0.7 • 1407
= 649400. kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.1416/2.0) • 600.0 • (44.0 - 3.0) • 0.74 • 1509
= 431523. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (64977 + 649400) = 714378 kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (64977 + 0 + 431523 + 0) = 496501 kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)



= (64977 + 431523 + 0) = 496501 kgf

Summary of Failure Path Calculations:

Path 1-1 = 714377 kgf, must exceed W = 339924 kgf or W1 = 156621 kgf
 Path 2-2 = 496500 kgf, must exceed W = 339924 kgf or W2 = 438710 kgf
 Path 3-3 = 496500 kgf, must exceed W = 339924 kgf or W3 = 438710 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 46 kgf/cm²

Nozzle is O.K. for the External Pressure: 1 kgf/cm²

The Drop for this Nozzle is : 16.7527 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 644.0000 mm

Input Echo, WRC107/537 Item 1, Description: N2600 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	5389.000	mm
Vessel Thickness	Tv	44.000	mm
Design Temperature	T1	275.0	C
Vessel Material		SA-387 11	
Vessel UNS Number		K11789	
Vessel Cold S.I. Allowable	Smc	1509.10	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1509.10	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	710.000	mm
Nozzle Thickness	Tn	75.000	mm
Nozzle Material		SA-182 F11	
Nozzle UNS Number		K11572	
Nozzle Cold S.I. Allowable	SNmc	1407.00	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1407.00	kgf/cm ²
Design Internal Pressure	Dp	44.531	kgf/cm ²
Include Pressure Thrust		No	

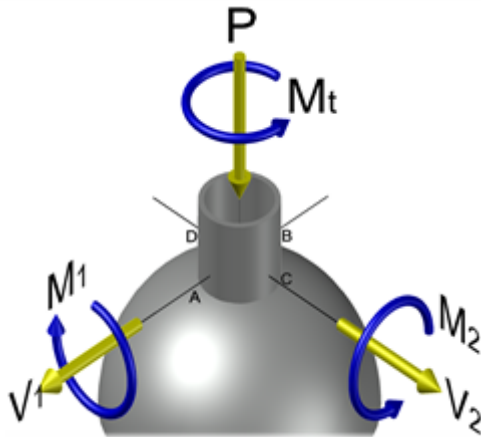
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	561.3	kgf
Longitudinal Shear (SUS)	(V1) V1	405.4	kgf
Circumferential Shear (SUS)	(Vc) V2	405.4	kgf
Circumferential Moment (SUS)	(Mc) M1	0.7	kgf-m.
Longitudinal Moment (SUS)	(Ml) M2	0.7	kgf-m.
Torsional Moment (SUS)	Mt	0.9	kgf-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 710.0 + 2 \cdot 1.65 \cdot \sqrt{(2718.0 (44.0 - 3.0))}$$

$$= 1811.617 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	561.3	kgf
Circumferential Shear	(VC) V2	405.4	kgf
Longitudinal Shear	(VL) V1	405.4	kgf
Circumferential Moment	(MC) M1	0.7	kgf-m.
Longitudinal Moment	(ML) M2	0.7	kgf-m.
Torsional Moment	MT	0.9	kgf-m.

Dimensionless Param: U = 1.06 TAU = 5.00 (4.43) RHO = 0.57

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.03816	(A,B,C,D)
$M(x) / P$	SP 1	0.03884	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.04562	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.06800	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.04562	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.06800	(A,B,C,D)
$N(y) * T / P$	SP 1	0.06352	(A,B,C,D)
$M(y) / P$	SP 1	0.01161	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.06288	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.02298	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.06288	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.02298	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	-1.3	-1.3	-1.3	-1.3	-1.3	-1.3	-1.3	-1.3
Rad. Bend.	P	-7.8	7.8	-7.8	7.8	-7.8	7.8	-7.8	7.8
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-0.1	0.1	0.1	-0.1
Rad. Memb.	ML	-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-0.1	0.1	0.1	-0.1	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-9.1	6.6	-9.0	6.5	-9.1	6.6	-9.0	6.5



Tang. Memb. P	-2.1	-2.1	-2.1	-2.1	-2.1	-2.1	-2.1	-2.1
Tang. Bend. P	-2.3	2.3	-2.3	2.3	-2.3	2.3	-2.3	2.3
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Tang. Bend. MC	0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Tang. Memb. ML	-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tang. Bend. ML	-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-4.5	0.2	-4.4	0.2	-4.5	0.2	-4.4	0.2
Shear VC	0.9	0.9	-0.9	-0.9	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.9	-0.9	0.9	0.9
Shear MT	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear	0.9	0.9	-0.9	-0.9	-0.9	-0.9	0.9	0.9
Str. Int.	9.3	6.7	9.2	6.6	9.3	6.7	9.2	6.6

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		1465.1	1465.1	1465.1	1465.1	1465.1	1465.1	1465.1	1465.1
Rad. Pl (SUS)		-1.3	-1.3	-1.3	-1.3	-1.3	-1.3	-1.3	-1.3
Rad. Q (SUS)		-7.8	7.8	-7.7	7.7	-7.8	7.8	-7.7	7.7
Long. Pm (SUS)		1465.1	1465.1	1465.1	1465.1	1465.1	1465.1	1465.1	1465.1
Long. Pl (SUS)		-2.1	-2.1	-2.1	-2.1	-2.1	-2.1	-2.1	-2.1
Long. Q (SUS)		-2.3	2.3	-2.3	2.3	-2.3	2.3	-2.3	2.3
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.9	0.9	-0.9	-0.9	-0.9	-0.9	0.9	0.9
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		1465.1	1465.1	1465.1	1465.1	1465.1	1465.1	1465.1	1465.1
Pm+Pl (SUS)		1464.4	1464.4	1464.4	1464.4	1464.4	1464.4	1464.4	1464.4
Pm+Pl+Q (Total)		1460.8	1471.8	1460.9	1471.7	1460.8	1471.8	1460.9	1471.7

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1465.13	1509.10	Passed
Pm+Pl (SUS)	1464.42	2263.65	Passed
Pm+Pl+Q (TOTAL)	1471.81	4527.30	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: K380

From: 40

Pressure for Reinforcement Calculations	P	44.531	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	275	°C

Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²

Inside Diameter of Hemispherical Head	D	5389.00	mm
Head Finished (Minimum) Thickness	t	44.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	1500.0000	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		80.0000	mm

Size and Thickness Basis		Minimum	
Nominal Thickness		160	

Flange Material		SA-182 F11	
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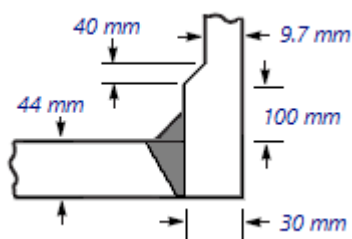
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	40.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	44.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.



Reinforcement CALCULATION, Description: K380

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	69.431 mm.
Actual Thickness Used in Calculation	9.735 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $44.53 \cdot 2697.5 / (2 \cdot 1509 \cdot 1.0 - 0.2 \cdot 44.53)$
= 39.9171 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $44.53 \cdot 37.72 / (1407 \cdot 1.0 - 0.6 \cdot 44.53)$
= 1.2168 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.6131 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
= 2.5 • Corroded Hub Thickness
= 2.5 • 27.0, less than the hub height, use (e-2)
= 67.5000 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

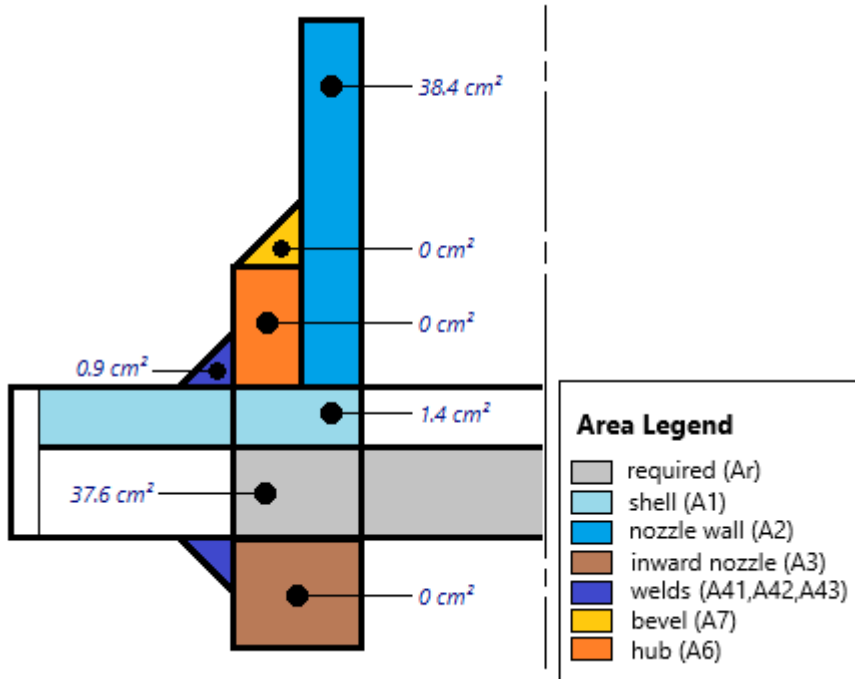
Parallel to Vessel Wall (Diameter Limit)	Dl	226.4704 mm
Parallel to Vessel Wall	Rn+tn+t	113.2352 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	67.5000 mm

Hub Height was ≥ 2.5 times Hub Thickness, using sketch UG-40 (e-2).

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n/S_v)$
= $\min(1, 1407.0/1509.1)$
= 0.932

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n/S_v)$
= $\min(1, 1407.0/1509.1)$
= 0.932

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(0.932, 1.0)$
= 0.932



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	37.572	3.867	NA
Area in Shell	A1	1.433	43.388	NA
Area in Nozzle Wall	A2	38.382	39.281	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.932	0.932	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	40.748	83.602	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 56.49 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (90.4704 \cdot 39.9171 \cdot 1.0 + 2 \cdot 27.0 \cdot 39.9171 \cdot 1.0 \cdot (1 - 0.93))$$

$$= 37.572 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 136.0(1.0 \cdot 41.0 - 1.0 \cdot 39.917) - 2 \cdot 27.0$$

$$(1.0 \cdot 41.0 - 1.0 \cdot 39.9171) \cdot (1 - 0.932)$$

$$= 1.433 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$$

$$= (2 \cdot 67.5)(27.0 - 1.22)0.9323/\sin(57.7)$$

$$= 38.382 \text{ cm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9323 + (0.0)^2 \cdot 0.9323$$



= 0.932 cm²

There are no hub area calculations because Figure UG-40 (e-2) was used.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 9.735, tr = 1.217, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.181$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 30.0, tr = 1.217, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.045$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-26 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $44.53/52.72 = 0.845$

Weld Size Calculations, Description: K380

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= $\max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
= $\max(0, (37.5715 - 1.4332 + 2 \cdot 27.0 \cdot 0.9323 \cdot (1.0 \cdot 41.0 - 39.9171)) \cdot 1509)$
= 55359.12 kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= $(A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
= $(38.3824 + 0.9323 - 0.0 \cdot 0.93) \cdot 1509$
= 59329.83 kgf

Weld Load [W2]:
= $(A2 + A3 + A4 + (2 \cdot (\text{Hub Thickness}) \cdot t \cdot fr1)) \cdot Sv$
= $(38.3824 + 0.0 + 0.9323 + (20.6421)) \cdot 1509$
= 90480.76 kgf

Weld Load [W3]:
= $(A2+A3+A4+(2 \cdot (\text{Hub Thickness}) \cdot t \cdot fr1)) \cdot Sv$
= $(38.3824 + 0.0 + 0.9323 + 0.0) \cdot 1509$
= 90480.76 kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 155.2371 \cdot 10.0 \cdot 0.49 \cdot 1407 \\ &= 16811. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 61.4269) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 1407 \\ &= 51317. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 155.2371 \cdot (44.0 - 3.0) \cdot 0.74 \cdot 1509 \\ &= 111647. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (16811 + 51317) = 68129 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (16811 + 0 + 111647 + 0) = 128459 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (16811 + 111647 + 0) = 128459 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 68128	kgf, must exceed W = 55359	kgf	or W1 = 59329	kgf
Path 2-2 = 128458	kgf, must exceed W = 55359	kgf	or W2 = 90480	kgf
Path 3-3 = 128458	kgf, must exceed W = 55359	kgf	or W3 = 90480	kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 46 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 kgf/cm²

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K380

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	69.431	mm.
Actual Thickness Used in Calculation	9.735	mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (2 \cdot Sv \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\ &= 44.53 \cdot 2697.5 / (2 \cdot 1509 \cdot 1.0 - 0.2 \cdot 44.53) \\ &= 39.9171 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned} &= P \cdot R / (Sn \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 44.53 \cdot 37.72 / (1407 \cdot 1.0 - 0.6 \cdot 44.53) \\ &= 1.2168 \text{ mm} \end{aligned}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.6131 mm

Intermediate Hub Nozzle Calculations:

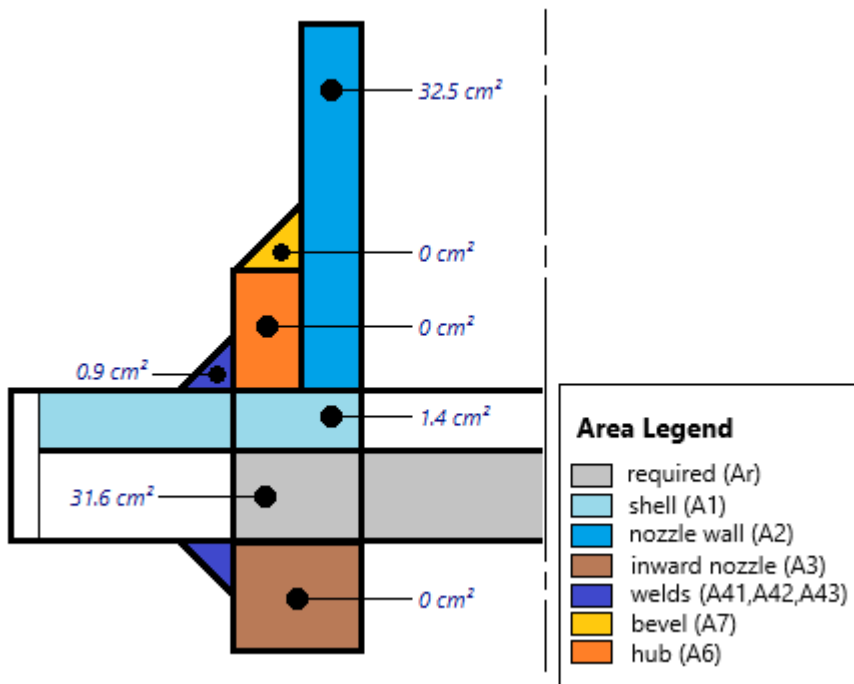
Check to determine use of Sketch (e-1) or (e-2):

$$\begin{aligned} &= 2.5 \cdot \text{Corroded Hub Thickness} \\ &= 2.5 \cdot 27.0, \text{ less than the hub height, use (e-2)} \\ &= 67.5000 \text{ mm} \end{aligned}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 211.4309 mm
 Parallel to Vessel Wall Rn+tn+t 105.7154 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 67.5000 mm

Hub Height was >= 2.5 times Hub Thickness, using sketch UG-40 (e-2).



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	31.568	3.249	NA
Area in Shell	A1	1.433	43.388	NA
Area in Nozzle Wall	A2	32.452	33.212	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.932	0.932	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	34.818	77.533	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (75.4309 \cdot 39.9171 \cdot 1.0 + 2 \cdot 27.0 \cdot 39.9171 \cdot 1.0 \cdot (1 - 0.93))$
 $= 31.568 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 136.0(1.0 \cdot 41.0 - 1.0 \cdot 39.9171) - 2 \cdot 27.0$
 $(1.0 \cdot 41.0 - 1.0 \cdot 39.9171) \cdot (1 - 0.932)$
 $= 1.433 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$



$$= (2 \cdot 67.5)(27.0 - 1.22) 0.9323$$
$$= 32.452 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i\text{-can}/0.707)^2 \cdot fr_2$$
$$= 10.0^2 \cdot 0.9323 + (0.0)^2 \cdot 0.9323$$
$$= 0.932 \text{ cm}^2$$

There are no hub area calculations because Figure UG-40 (e-2) was used.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.2168 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 42.9171 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 42.9171 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.9426 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$
$$= \min[7.8, \max(42.9171, 4.5)]$$
$$= 7.8000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$
$$= \max(4.2168, 7.8)$$
$$= 7.8000 \text{ mm}$$

Available Nozzle Neck Thickness = 9.7346 mm -> OK

Weld Size Calculations, Description: K380

Intermediate Calc. for nozzle/shell welds	Tmin	19.0000 mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$
$$= \max(0, (31.5682 - 1.4332 + 2 \cdot 27.0 \cdot 0.9323 \cdot (1.0 \cdot 41.0 - 39.9171)) \cdot 1509)$$
$$= 46299.48 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i\text{-Can}/.707)^2 \cdot fr_2) \cdot Sv$$
$$= (32.4524 + 0.9323 - 0.0 \cdot 0.93) \cdot 1509$$
$$= 50380.85 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot (\text{Hub Thickness}) \cdot t \cdot fr_1)) \cdot Sv$$
$$= (32.4524 + 0.0 + 0.9323 + (20.6421)) \cdot 1509$$
$$= 81531.79 \text{ kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+(2 \cdot (\text{Hub Thickness}) \cdot t \cdot fr_1)) \cdot Sv$$
$$= (32.4524 + 0.0 + 0.9323 + 0.0) \cdot 1509$$
$$= 81531.79 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_o \cdot W_o \cdot 0.49 \cdot S_{nw}$$



$$= (3.1416/2.0) \cdot 129.4309 \cdot 10.0 \cdot 0.49 \cdot 1407$$
$$= 14017. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$
$$= (3.1416 \cdot 51.2155) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 1407$$
$$= 42786. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$
$$= (3.1416/2.0) \cdot 129.4309 \cdot (44.0 - 3.0) \cdot 0.74 \cdot 1509$$
$$= 93087. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (14017 + 42786) = 56803 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$
$$= (14017 + 0 + 93087 + 0) = 107104 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$
$$= (14017 + 93087 + 0) = 107104 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 56803	kgf, must exceed W = 46299 kgf	or W1 = 50380 kgf
Path 2-2 = 107104	kgf, must exceed W = 46299 kgf	or W2 = 81531 kgf
Path 3-3 = 107104	kgf, must exceed W = 46299 kgf	or W3 = 81531 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 46 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 kgf/cm²

The Drop for this Nozzle is : 44.7504 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 597.5268 mm

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Input, Nozzle Desc: UC-1

From: 50

Pressure for Reinforcement Calculations	P	44.531	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	275	°C

Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	5350.00	mm
Design Length of Section	L	4696.3335	mm
Shell Finished (Minimum) Thickness	t	84.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		90.00	cm
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User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		600.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm

Flange Material		SA-182 F11	
-----------------	--	------------	--

Hub Height of Integral Nozzle	h	160.0000	mm
Height of Beveled Transition	L'	126.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		160.0000	mm

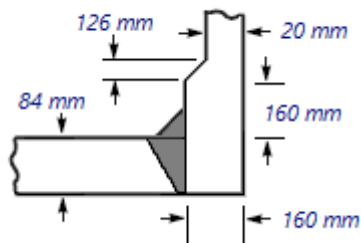
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	84.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

This is a Manway, Coupling or Access Opening.

Flange Class		300	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: UC-1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	600.000 mm.
Actual Thickness Used in Calculation	20.000 mm

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 44.53 \cdot 2678.0 / (1509 \cdot 1.0 - 0.6 \cdot 44.53)$
 $= 80.4476$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 44.53 \cdot 303.0 / (1407 \cdot 1.0 - 0.6 \cdot 44.53)$
 $= 9.7755$ mm

Required Nozzle thickness under External Pressure per UG-28 : 1.6518 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 140.0 / 0.5773$
 $= 242.4873$ mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

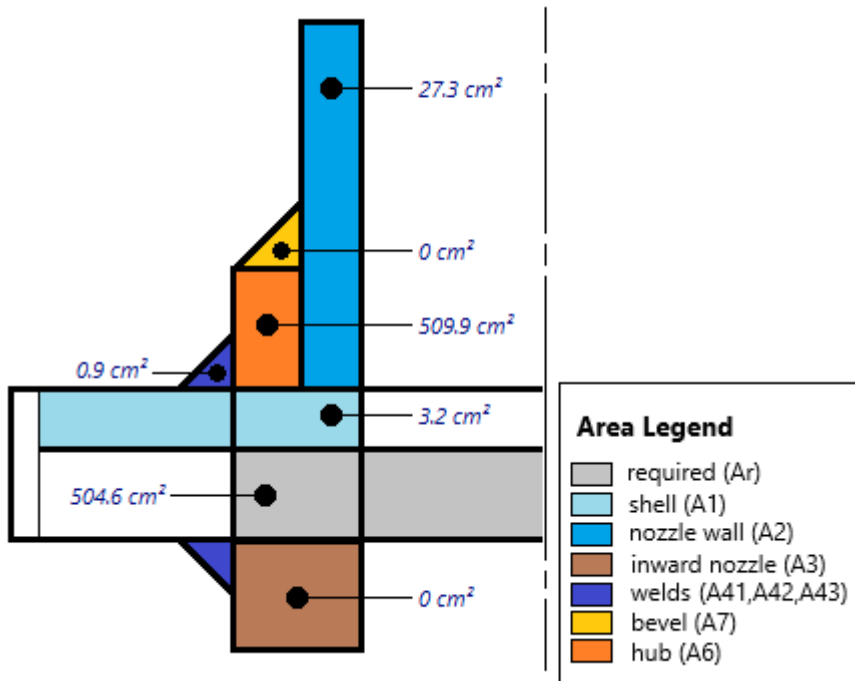
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1212.0000 mm
Parallel to Vessel Wall, opening length	d	606.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	202.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407.0 / 1509.1)$
 $= 0.932$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407.0 / 1509.1)$
 $= 0.932$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.932, 1.0)$
 $= 0.932$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	504.603	53.575	NA
Area in Shell	A1	3.230	373.761	NA
Area in Nozzle Wall	A2	27.280	57.955	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.932	0.932	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	509.927	509.927	NA
TOTAL AREA AVAILABLE	Atot	541.370	942.575	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (606.0 \cdot 80.4476 \cdot 1.0 + 2 \cdot 157.0 \cdot 80.4476 \cdot 1.0 \cdot (1 - 0.93))$
 $= 504.603 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 606.0(1.0 \cdot 81.0 - 1.0 \cdot 80.448) - 2 \cdot 157.0$
 $(1.0 \cdot 81.0 - 1.0 \cdot 80.4476) \cdot (1 - 0.932)$
 $= 3.230 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 202.5)(17.0 - 9.78)0.9323$
 $= 27.280 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9323 + (0.0)^2 \cdot 0.9323$
 $= 0.932 \text{ cm}^2$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2 \\ &= (2 \cdot \min(202.5, 350.0, 160.0)) \cdot (160.0 - 20.0) \cdot 0.932 \\ &= 509.927 \text{ cm}^2 \text{ Includes Bevel Area of } 92.237 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 12.7755 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= t_a \\ &= 12.7755 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:A)

Govrn. thk, $t_g = 84.0$, $t_r = 80.448$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.993$, Temp. Reduction = $0 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	46 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	45 $^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : 45 $^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 $^\circ\text{C}$
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-26 $^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $44.53/52.72 = 0.845$

Weld Size Calculations, Description: UC-1

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (504.6027 - 3.2304 + 2 \cdot 157.0 \cdot 0.9323 \cdot (1.0 \cdot 81.0 - 80.4476)) \cdot 1509)$
 $= 759060.62 \text{ kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (537.207 + 0.9323 - 0.0 \cdot 0.93) \cdot 1509$
 $= 812105.25 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (537.207 + 0.0 + 0.9323 + (237.1322)) \cdot 1509$
 $= 1169961.12 \text{ kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (537.207 + 0.0 + 0.9323 + 0.0 + (237.1322)) \cdot 1509$
 $= 1169961.12 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 920.0 \cdot 10.0 \cdot 0.49 \cdot 1407 \\ &= 99632. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 381.5) \cdot (160.0 - 3.0) \cdot 0.7 \cdot 1407 \\ &= 1853257. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 920.0 \cdot (84.0 - 3.0) \cdot 0.74 \cdot 1509 \\ &= 1307200. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (99632 + 1853257) = 1952889 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (99632 + 0 + 1307200 + 0) = 1406832 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (99632 + 1307200 + 0) = 1406832 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 1952888 kgf, must exceed W = 759060 kgf or W1 = 812105 kgf
Path 2-2 = 1406831 kgf, must exceed W = 759060 kgf or W2 = 1169961 kgf
Path 3-3 = 1406831 kgf, must exceed W = 759060 kgf or W3 = 1169961 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 45 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 kgf/cm²

The Drop for this Nozzle is : 39.8481 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 473.8481 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 606.00 mm

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Input, Nozzle Desc: M1600

From: 60

Pressure for Reinforcement Calculations	P	44.531	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	275	°C

Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²

Inside Diameter of Hemispherical Head	D	5389.00	mm
Head Finished (Minimum) Thickness	t	44.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	1500.0000	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		600.0000	mm

Size and Thickness Basis		Minimum	
Nominal Thickness		160	

Flange Material		SA-182 F11	
-----------------	--	------------	--

Hub Height of Integral Nozzle	h	200.0000	mm
Height of Beveled Transition	L'	164.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		160.0000	mm

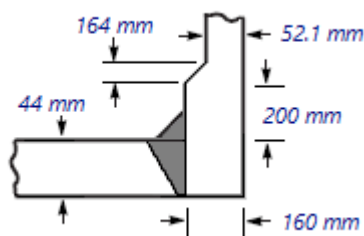
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	1000.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	44.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

This is a Manway, Coupling or Access Opening.

Flange Class		300	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.





Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: M1600

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	505.454 mm.
Actual Thickness Used in Calculation	52.073 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $44.53 \cdot 2697.5 / (2 \cdot 1509 \cdot 1.0 - 0.2 \cdot 44.53)$
= 39.9171 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $44.53 \cdot 255.73 / (1407 \cdot 1.0 - 0.6 \cdot 44.53)$
= 8.2503 mm

Required Nozzle thickness under External Pressure per UG-28 : 2.4857 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 107.927/0.5773
= 186.9349 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

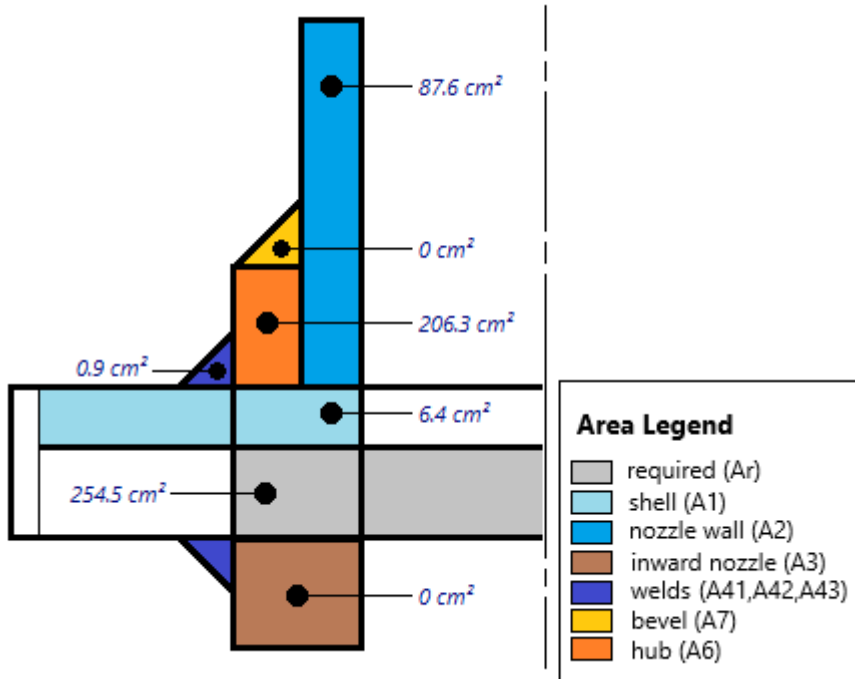
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1232.8678 mm
Parallel to Vessel Wall, opening length	d	616.4339 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	102.5000 mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 1407.0/1509.1)
= 0.932

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 1407.0/1509.1)
= 0.932

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.932, 1.0)
= 0.932



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	254.543	26.197	NA
Area in Shell	A1	6.445	195.125	NA
Area in Nozzle Wall	A2	87.588	99.956	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.932	0.932	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	206.281	206.281	NA
TOTAL AREA AVAILABLE	Atot	301.247	502.295	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 56.07 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (616.4339 \cdot 39.9171 \cdot 1.0 + 2 \cdot 157.0 \cdot 39.9171 \cdot 1.0 \cdot (1 - 0.93))$
 $= 254.543 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 616.434(1.0 \cdot 41.0 - 1.0 \cdot 39.917) - 2 \cdot 157.0$
 $(1.0 \cdot 41.0 - 1.0 \cdot 39.9171) \cdot (1 - 0.932)$
 $= 6.445 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 102.5)(49.07 - 8.25)0.9323/\sin(63.0)$
 $= 87.588 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9323 + (0.0)^2 \cdot 0.9323$



$$= 0.932 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2 \\ &= (2 \cdot \min(102.5, 1000.0, 200.0)) \cdot (160.0 - 52.1) \cdot 0.932 \\ &= 206.281 \text{ cm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 52.073, tr = 8.25, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.168, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 44.0, tr = 39.917, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.974, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	33 °C
Min Metal Temp. at Required thickness (UCS 66.1)	32 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-26 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 44.53/52.72 = 0.845

Weld Size Calculations, Description: M1600

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A-A1+2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv) \\ &= \max(0, (254.5427 - 6.4452 + 2 \cdot 157.0 \cdot 0.9323 \cdot \\ &\quad (1.0 \cdot 41.0 - 39.9171)) \cdot 1509) \\ &= 379187.62 \text{ kgf} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A4-(W_i \cdot Can/.707)^2 \cdot fr2) \cdot Sv \\ &= (293.869 + 0.9323 - 0.0 \cdot 0.93) \cdot 1509 \\ &= 444884.19 \text{ kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv \\ &= (293.869 + 0.0 + 0.9323 + (120.0299)) \cdot 1509 \\ &= 626021.06 \text{ kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S \\ &= (293.869 + 0.0 + 0.9323 + 0.0 + (120.0299)) \cdot 1509 \\ &= 626021.06 \text{ kgf} \end{aligned}$$



Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 994.8851 \cdot 10.0 \cdot 0.49 \cdot 1407 \\ &= 107741. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 402.8297) \cdot (160.0 - 3.0) \cdot 0.7 \cdot 1407 \\ &= 1956873. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 994.8851 \cdot (44.0 - 3.0) \cdot 0.74 \cdot 1509 \\ &= 715527. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (107741 + 1956873) = 2064614 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (107741 + 0 + 715527 + 0) = 823268 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (107741 + 715527 + 0) = 823268 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 2064614 kgf, must exceed W = 379187 kgf	or W1 = 444884 kgf
Path 2-2 = 823268 kgf, must exceed W = 379187 kgf	or W2 = 626021 kgf
Path 3-3 = 823268 kgf, must exceed W = 379187 kgf	or W3 = 626021 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 46 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 kgf/cm²

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: M1600

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	505.454 mm.
Actual Thickness Used in Calculation	52.073 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\ &= 44.53 \cdot 2697.5 / (2 \cdot 1509 \cdot 1.0 - 0.2 \cdot 44.53) \\ &= 39.9171 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned} &= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 44.53 \cdot 255.73 / (1407 \cdot 1.0 - 0.6 \cdot 44.53) \\ &= 8.2503 \text{ mm} \end{aligned}$$

Required Nozzle thickness under External Pressure per UG-28 : 2.4857 mm

Intermediate Hub Nozzle Calculations:

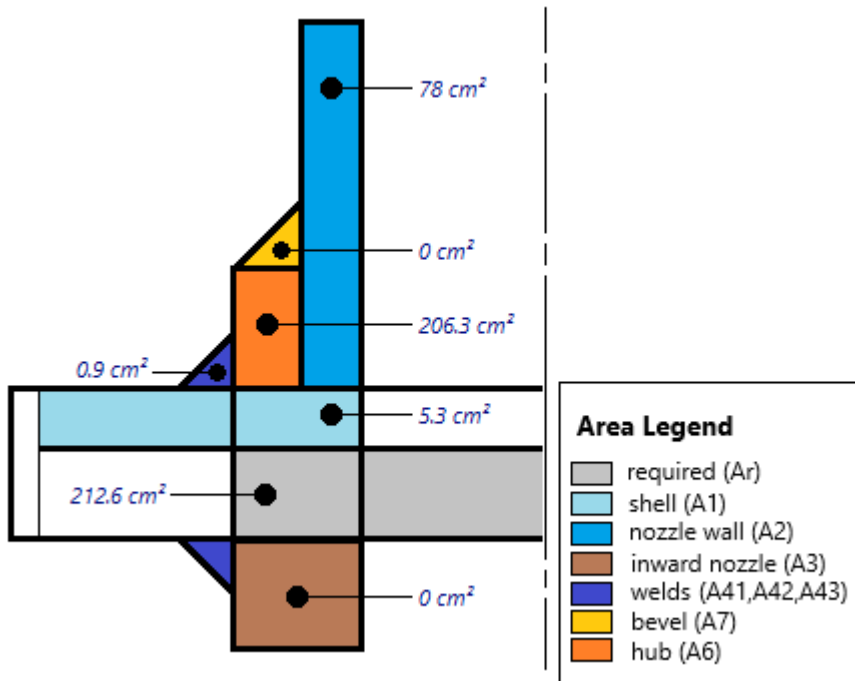
Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 107.927/0.5773$
 $= 186.9349 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1022.9073 mm
 Parallel to Vessel Wall, opening length d 511.4536 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 102.5000 mm



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	212.638	21.884	NA
Area in Shell	A1	5.308	160.709	NA
Area in Nozzle Wall	A2	78.025	89.043	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		0.932	0.932	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	206.281	206.281	NA
TOTAL AREA AVAILABLE	Atot	290.547	456.965	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (511.4536 \cdot 39.9171 \cdot 1.0 + 2 \cdot 157.0 \cdot 39.9171 \cdot 1.0 \cdot (1 - 0.93))$
 $= 212.638 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 511.454(1.0 \cdot 41.0 - 1.0 \cdot 39.917) - 2 \cdot 157.0$



$$(1.0 \cdot 41.0 - 1.0 \cdot 39.9171) \cdot (1 - 0.932) \\ = 5.308 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) \cdot fr_2 \\ = (2 \cdot 102.5) (49.07 - 8.25) \cdot 0.9323 \\ = 78.025 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2 \\ = 10.0^2 \cdot 0.9323 + (0.0)^2 \cdot 0.9323 \\ = 0.932 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr_2 \\ = (2 \cdot \min(102.5, 1000.0, 200.0)) \cdot (160.0 - 52.1) \cdot 0.932 \\ = 206.281 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 11.2503 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a \\ = 11.2503 \text{ mm}$$

Available Nozzle Neck Thickness = 52.0732 mm -> OK

Weld Size Calculations, Description: M1600

Intermediate Calc. for nozzle/shell welds $T_{min} = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr)) \cdot S_v) \\ = \max(0, (212.6376 - 5.3084 + 2 \cdot 157.0 \cdot 0.9323 \cdot (1.0 \cdot 41.0 - 39.9171)) \cdot 1509) \\ = 317664.28 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - can/0.707)^2 \cdot fr_2) \cdot S_v \\ = (284.3058 + 0.9323 - 0.0 \cdot 0.93) \cdot 1509 \\ = 430452.47 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v \\ = (284.3058 + 0.0 + 0.9323 + (120.0299)) \cdot 1509 \\ = 611589.31 \text{ kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S \\ = (284.3058 + 0.0 + 0.9323 + 0.0 + (120.0299)) \cdot 1509 \\ = 611589.31 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [S_{onw}]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ = (3.1416/2.0) \cdot 825.4537 \cdot 10.0 \cdot 0.49 \cdot 1407 \\ = 89393. \text{ kgf}$$

Shear, Nozzle Wall [S_{nw}]:



$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 334.2268) \cdot (160.0 - 3.0) \cdot 0.7 \cdot 1407 \\ &= 1623612. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 825.4537 \cdot (44.0 - 3.0) \cdot 0.74 \cdot 1509 \\ &= 593671. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (89393 + 1623613) = 1713005 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (89393 + 0 + 593671 + 0) = 683064 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (89393 + 593671 + 0) = 683064 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 1713005 kgf, must exceed W = 317664 kgf or W1 = 430452 kgf
Path 2-2 = 683063 kgf, must exceed W = 317664 kgf or W2 = 611589 kgf
Path 3-3 = 683063 kgf, must exceed W = 317664 kgf or W3 = 611589 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 46 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 kgf/cm²

The Drop for this Nozzle is : 340.5341 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 1393.3104 mm

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Input, Nozzle Desc: K150

From: 60

Pressure for Reinforcement Calculations	P	44.531	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	275	°C

Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²

Inside Diameter of Hemispherical Head	D	5389.00	mm
Head Finished (Minimum) Thickness	t	44.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	1500.0000	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		240.00	deg
Diameter		50.0000	mm

Size and Thickness Basis		Minimum	
Nominal Thickness		XXS	

Flange Material		SA-182 F11	
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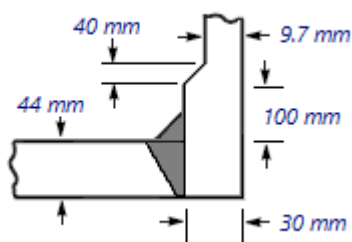
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	40.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	1000.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	44.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.



Reinforcement CALCULATION, Description: K150

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	40.945	mm.
Actual Thickness Used in Calculation	9.690	mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $44.53 \cdot 2697.5 / (2 \cdot 1509 \cdot 1.0 - 0.2 \cdot 44.53)$
= 39.9171 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $44.53 \cdot 23.47 / (1407 \cdot 1.0 - 0.6 \cdot 44.53)$
= 0.7573 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.5438 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
= 2.5 • Corroded Hub Thickness
= 2.5 • 27.0, less than the hub height, use (e-2)
= 67.5000 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

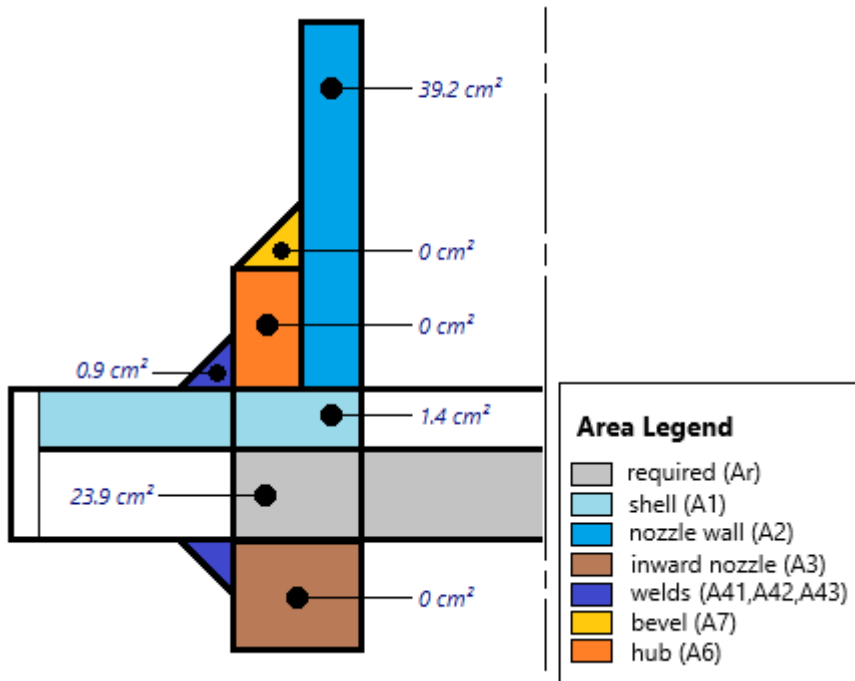
Parallel to Vessel Wall (Diameter Limit)	Dl	192.3011	mm
Parallel to Vessel Wall	Rn+tn+t	96.1505	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	67.5000	mm

Hub Height was ≥ 2.5 times Hub Thickness, using sketch UG-40 (e-2).

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, $1407.0/1509.1$)
= 0.932

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, $1407.0/1509.1$)
= 0.932

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.932, 1.0)
= 0.932



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	23.932	2.463	NA
Area in Shell	A1	1.433	43.388	NA
Area in Nozzle Wall	A2	39.220	39.539	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.932	0.932	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	41.585	83.859	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 56.49 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.3011 \cdot 39.9171 \cdot 1.0 + 2 \cdot 27.0 \cdot 39.9171 \cdot 1.0 \cdot (1 - 0.93))$
 $= 23.932 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 136.0(1.0 \cdot 41.0 - 1.0 \cdot 39.917) - 2 \cdot 27.0$
 $(1.0 \cdot 41.0 - 1.0 \cdot 39.9171) \cdot (1 - 0.932)$
 $= 1.433 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 67.5)(27.0 - 0.76)0.9323/\sin(57.4)$
 $= 39.220 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9323 + (0.0)^2 \cdot 0.9323$



= 0.932 cm²

There are no hub area calculations because Figure UG-40 (e-2) was used.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 9.69, tr = 0.757, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.113$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 30.0, tr = 0.757, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.028$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-26 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $44.53/52.72 = 0.845$

Weld Size Calculations, Description: K150

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= $\max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
= $\max(0, (23.9321 - 1.4332 + 2 \cdot 27.0 \cdot 0.9323 \cdot (1.0 \cdot 41.0 - 39.9171)) \cdot 1509)$
= 34775.89 kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= $(A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
= $(39.2199 + 0.9323 - 0.0 \cdot 0.93) \cdot 1509$
= 60593.66 kgf

Weld Load [W2]:
= $(A2 + A3 + A4 + (2 \cdot (\text{Hub Thickness}) \cdot t \cdot fr1)) \cdot Sv$
= $(39.2199 + 0.0 + 0.9323 + (20.6421)) \cdot 1509$
= 91744.59 kgf

Weld Load [W3]:
= $(A2+A3+A4+(2 \cdot (\text{Hub Thickness}) \cdot t \cdot fr1)) \cdot Sv$
= $(39.2199 + 0.0 + 0.9323 + 0.0) \cdot 1509$
= 91744.59 kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 121.0635 \cdot 10.0 \cdot 0.49 \cdot 1407 \\ &= 13111. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 44.3411) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 1407 \\ &= 37043. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 121.0635 \cdot (44.0 - 3.0) \cdot 0.74 \cdot 1509 \\ &= 87070. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (13111 + 37043) = 50154 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (13111 + 0 + 87070 + 0) = 100180 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (13111 + 87070 + 0) = 100180 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 50154	kgf, must exceed W = 34775 kgf	or W1 = 60593 kgf
Path 2-2 = 100180	kgf, must exceed W = 34775 kgf	or W2 = 91744 kgf
Path 3-3 = 100180	kgf, must exceed W = 34775 kgf	or W3 = 91744 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 46 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 kgf/cm²

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K150

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	40.945 mm.
Actual Thickness Used in Calculation	9.690 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (2 \cdot Sv \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\ &= 44.53 \cdot 2697.5 / (2 \cdot 1509 \cdot 1.0 - 0.2 \cdot 44.53) \\ &= 39.9171 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned} &= P \cdot R / (Sn \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 44.53 \cdot 23.47 / (1407 \cdot 1.0 - 0.6 \cdot 44.53) \\ &= 0.7573 \text{ mm} \end{aligned}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.5438 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

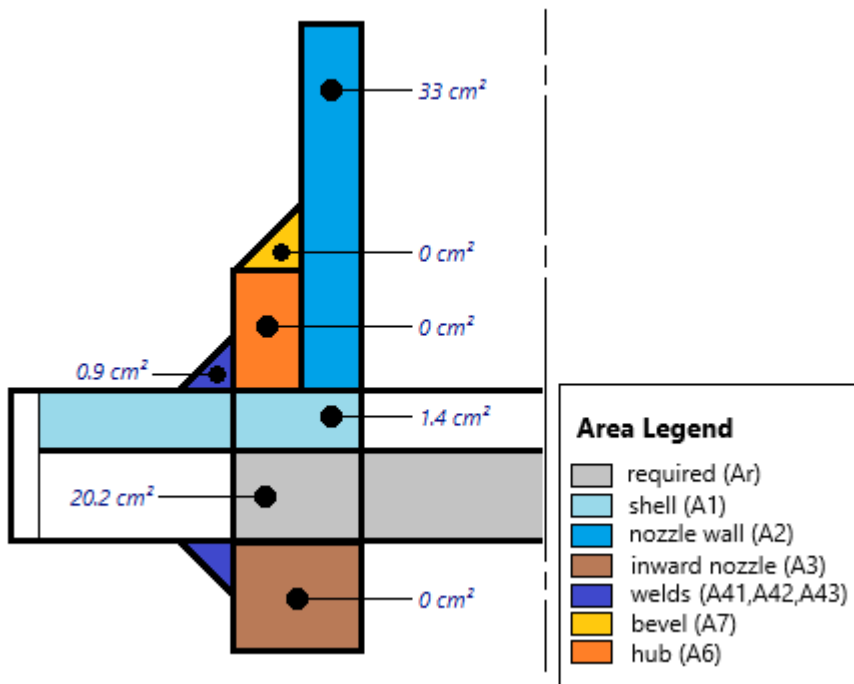
$$\begin{aligned} &= 2.5 \cdot \text{Corroded Hub Thickness} \\ &= 2.5 \cdot 27.0, \text{ less than the hub height, use (e-2)} \\ &= 67.5000 \text{ mm} \end{aligned}$$



UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 182.9448 mm
 Parallel to Vessel Wall Rn+tn+t 91.4724 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 67.5000 mm

Hub Height was >= 2.5 times Hub Thickness, using sketch UG-40 (e-2).



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	20.197	2.079	NA
Area in Shell	A1	1.433	43.388	NA
Area in Nozzle Wall	A2	33.031	33.299	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.932	0.932	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	35.396	77.620	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (46.9448 \cdot 39.9171 \cdot 1.0 + 2 \cdot 27.0 \cdot 39.9171 \cdot 1.0 \cdot (1 - 0.93))$
 $= 20.197 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 136.0(1.0 \cdot 41.0 - 1.0 \cdot 39.917) - 2 \cdot 27.0$
 $(1.0 \cdot 41.0 - 1.0 \cdot 39.9171) \cdot (1 - 0.932)$
 $= 1.433 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$



$$= (2 \cdot 67.5)(27.0 - 0.76) 0.9323$$

$$= 33.031 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 0.9323 + (0.0)^2 \cdot 0.9323$$

$$= 0.932 \text{ cm}^2$$

There are no hub area calculations because Figure UG-40 (e-2) was used.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.7573 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 42.9171 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 42.9171 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.9426 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[6.42, \max(42.9171, 4.5)]$$

$$= 6.4200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.7573, 6.42)$$

$$= 6.4200 \text{ mm}$$

Available Nozzle Neck Thickness = 9.6901 mm -> OK

Weld Size Calculations, Description: K150

Intermediate Calc. for nozzle/shell welds	Tmin	19.0000 mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (20.1974 - 1.4332 + 2 \cdot 27.0 \cdot 0.9323 \cdot (1.0 \cdot 41.0 - 39.9171)) \cdot 1509)$$

$$= 29139.79 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - Can/0.707)^2 \cdot fr_2) \cdot Sv$$

$$= (33.0308 + 0.9323 - 0.0 \cdot 0.93) \cdot 1509$$

$$= 51253.68 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot (\text{Hub Thickness}) \cdot t \cdot fr1)) \cdot Sv$$

$$= (33.0308 + 0.0 + 0.9323 + (20.6421)) \cdot 1509$$

$$= 82404.61 \text{ kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + (2 \cdot (\text{Hub Thickness}) \cdot t \cdot fr1)) \cdot Sv$$

$$= (33.0308 + 0.0 + 0.9323 + 0.0) \cdot 1509$$

$$= 82404.61 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_o \cdot W_o \cdot 0.49 \cdot Snw$$



$$= (3.1416/2.0) \cdot 100.9448 \cdot 10.0 \cdot 0.49 \cdot 1407$$
$$= 10932. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$
$$= (3.1416 \cdot 36.9724) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 1407$$
$$= 30888. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$
$$= (3.1416/2.0) \cdot 100.9448 \cdot (44.0 - 3.0) \cdot 0.74 \cdot 1509$$
$$= 72600. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (10932 + 30888) = 41819 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$
$$= (10932 + 0 + 72600 + 0) = 83532 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$
$$= (10932 + 72600 + 0) = 83532 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 41819	kgf, must exceed W = 29139 kgf	or W1 = 51253 kgf
Path 2-2 = 83531	kgf, must exceed W = 29139 kgf	or W2 = 82404 kgf
Path 3-3 = 83531	kgf, must exceed W = 29139 kgf	or W3 = 82404 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 46 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 kgf/cm²

The Drop for this Nozzle is : 34.6603 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1087.4368 mm

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Input, Nozzle Desc: N1600

From: 60

Pressure for Reinforcement Calculations	P	44.531	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	275	°C

Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²

Inside Diameter of Hemispherical Head	D	5389.00	mm
Head Finished (Minimum) Thickness	t	44.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		600.0000	mm

Size and Thickness Basis		Minimum	
Nominal Thickness		80	

Flange Material		SA-182 F11	
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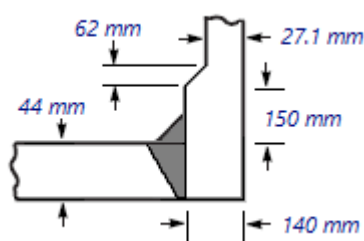
Hub Height of Integral Nozzle	h	150.0000	mm
Height of Beveled Transition	L'	62.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		140.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	1000.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	44.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.





ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 555.460 mm.
Actual Thickness Used in Calculation 27.070 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $44.53 \cdot 2697.5 / (2 \cdot 1509 \cdot 1.0 - 0.2 \cdot 44.53)$
= 39.9171 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $44.53 \cdot 280.73 / (1407 \cdot 1.0 - 0.6 \cdot 44.53)$
= 9.0570 mm

Required Nozzle thickness under External Pressure per UG-28 : 2.4857 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 112.93/0.5773
= 195.6005 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

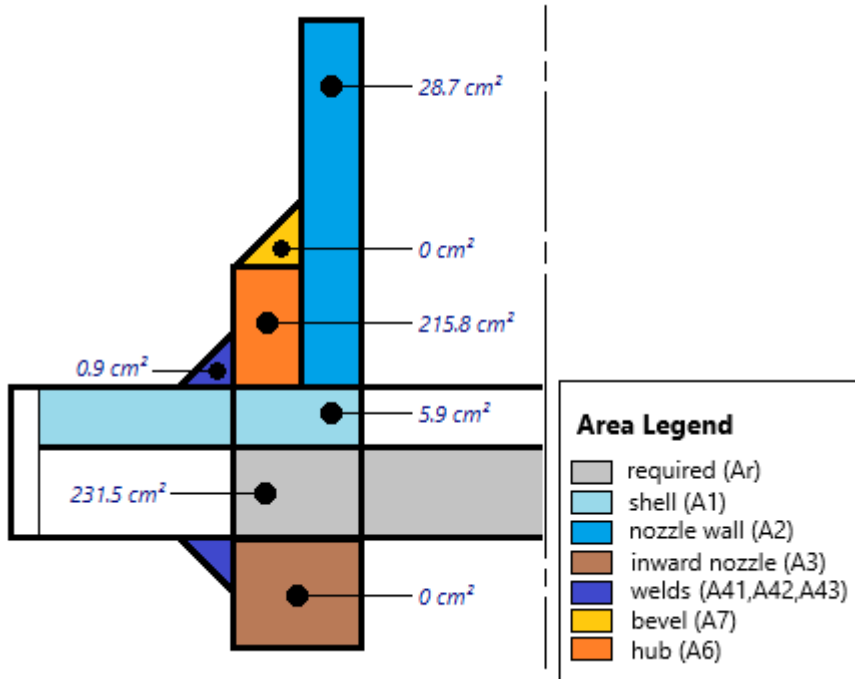
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	1122.9198	mm
Parallel to Vessel Wall, opening length	d	561.4599	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	102.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 1407.0/1509.1)
= 0.932

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 1407.0/1509.1)
= 0.932

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.932, 1.0)
= 0.932



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	231.518	23.827	NA
Area in Shell	A1	5.879	177.990	NA
Area in Nozzle Wall	A2	28.695	41.254	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.932	0.932	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	215.843	215.843	NA
TOTAL AREA AVAILABLE	Atot	251.350	436.020	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (561.4599 \cdot 39.9171 \cdot 1.0 + 2 \cdot 137.0 \cdot 39.9171 \cdot 1.0 \cdot (1 - 0.93))$$

$$= 231.518 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 561.46(1.0 \cdot 41.0 - 1.0 \cdot 39.917) - 2 \cdot 137.0$$

$$(1.0 \cdot 41.0 - 1.0 \cdot 39.9171) \cdot (1 - 0.932)$$

$$= 5.879 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 102.5)(24.07 - 9.06) \cdot 0.9323$$

$$= 28.695 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9323 + (0.0)^2 \cdot 0.9323$$

$$= 0.932 \text{ cm}^2$$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2 \\ &= (2 \cdot \min(102.5, 1000.0, 150.0)) \cdot (140.0 - 27.1) \cdot 0.932 \\ &= 215.843 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 12.0570 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 42.9171 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 42.9171 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.9426 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[11.331, \max(42.9171, 4.5)] \\ &= 11.3312 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(12.057, 11.3312) \\ &= 12.0570 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 27.0701 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 250.6,	Allowable	: 1407.0 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3266.9 kgf/cm ²	Passed
Occasional	: 249.0,	Allowable	: 1871.3 kgf/cm ²	Passed
Shear	: 3.4,	Allowable	: 984.9 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 27.07, tr = 9.057, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.376, Temp. Reduction = 58 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 44.0, tr = 39.917, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.974, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	33 °C
Min Metal Temp. at Required thickness (UCS 66.1)	32 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-26 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 44.53/52.72 = 0.845

Weld Size Calculations, Description: N1600

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:



	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (231.5184 - 5.8792 + 2 \cdot 137.0 \cdot 0.9323 \cdot (1.0 \cdot 41.0 - 39.9171)) \cdot 1509)$
 $= 344686.44 \text{ kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (244.538 + 0.9323 - 0.0 \cdot 0.93) \cdot 1509$
 $= 370438.91 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (244.538 + 0.0 + 0.9323 + (104.7395)) \cdot 1509$
 $= 528501.06 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (244.538 + 0.0 + 0.9323 + 0.0 + (104.7395)) \cdot 1509$
 $= 528501.06 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 835.4598 \cdot 10.0 \cdot 0.49 \cdot 1407$
 $= 90476. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 349.2299) \cdot (140.0 - 3.0) \cdot 0.7 \cdot 1407$
 $= 1480381. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 835.4598 \cdot (44.0 - 3.0) \cdot 0.74 \cdot 1509$
 $= 600867. \text{ kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (90476 + 1480381) = 1570857 \text{ kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (90476 + 0 + 600867 + 0) = 691344 \text{ kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (90476 + 600867 + 0) = 691344 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 1570857 kgf, must exceed W = 344686 kgf	or W1 = 370438 kgf
Path 2-2 = 691343 kgf, must exceed W = 344686 kgf	or W2 = 528501 kgf
Path 3-3 = 691343 kgf, must exceed W = 344686 kgf	or W3 = 528501 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 46 kgf/cm²



Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 kgf/cm²

The Drop for this Nozzle is : 32.5773 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1076.5773 mm

Input Echo, WRC107/537 Item 1, Description: N1600 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	5389.000	mm
Vessel Thickness	Tv	44.000	mm
Design Temperature	T1	275.0	C
Vessel Material		SA-387 11	
Vessel UNS Number		K11789	
Vessel Cold S.I. Allowable	Smc	1509.10	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1509.10	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	555.460	mm
Nozzle Thickness	Tn	140.000	mm
Nozzle Material		SA-182 F11	
Nozzle UNS Number		K11572	
Nozzle Cold S.I. Allowable	SNmc	1407.00	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1407.00	kgf/cm ²
Design Internal Pressure	Dp	44.531	kgf/cm ²
Include Pressure Thrust		No	

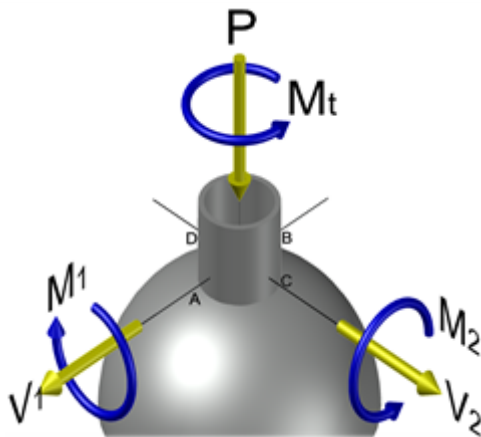
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	701.7	kgf
Longitudinal Shear (SUS)	(V1) V1	506.8	kgf
Circumferential Shear (SUS)	(Vc) V2	506.8	kgf
Circumferential Moment (SUS)	(Mc) M1	1.2	kgf-m.
Longitudinal Moment (SUS)	(Ml) M2	1.2	kgf-m.
Torsional Moment (SUS)	Mt	1.3	kgf-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 835.46 + 2 \cdot 1.65 \cdot \sqrt{(2718.0 (44.0 - 3.0))}$$

$$= 1937.077 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	701.7	kgf
Circumferential Shear	(VC) V2	506.8	kgf
Longitudinal Shear	(VL) V1	506.8	kgf
Circumferential Moment	(MC) M1	1.2	kgf-m.
Longitudinal Moment	(ML) M2	1.2	kgf-m.
Torsional Moment	MT	1.3	kgf-m.

Dimensionless Param: $U = 1.25 \text{ TAU} = 5.00 (2.55) \text{ RHO} = 0.30$

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.03195	(A,B,C,D)
$M(x) / P$	SP 1	0.03299	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.03314	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.05269	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.03314	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.05269	(A,B,C,D)
$N(y) * T / P$	SP 1	0.04732	(A,B,C,D)
$M(y) / P$	SP 1	0.00901	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.04530	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.01528	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.04530	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.01528	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	-1.3	-1.3	-1.3	-1.3	-1.3	-1.3	-1.3	-1.3
Rad. Bend.	P	-8.3	8.3	-8.3	8.3	-8.3	8.3	-8.3	8.3
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-0.1	0.1	0.1	-0.1
Rad. Memb.	ML	-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-0.1	0.1	0.1	-0.1	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-9.7	7.0	-9.5	6.9	-9.7	7.0	-9.5	6.9



Tang. Memb. P	-2.0	-2.0	-2.0	-2.0	-2.0	-2.0	-2.0	-2.0
Tang. Bend. P	-2.3	2.3	-2.3	2.3	-2.3	2.3	-2.3	2.3
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Tang. Bend. MC	0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Tang. Memb. ML	-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tang. Bend. ML	-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-4.3	0.3	-4.2	0.3	-4.3	0.3	-4.2	0.3
Shear VC	0.9	0.9	-0.9	-0.9	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.9	-0.9	0.9	0.9
Shear MT	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear	0.9	0.9	-0.9	-0.9	-0.9	-0.9	0.9	0.9
Str. Int.	9.8	7.1	9.7	7.0	9.8	7.1	9.7	7.0

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		1465.1	1465.1	1465.1	1465.1	1465.1	1465.1	1465.1	1465.1
Rad. Pl (SUS)		-1.3	-1.3	-1.3	-1.3	-1.3	-1.3	-1.3	-1.3
Rad. Q (SUS)		-8.3	8.3	-8.2	8.2	-8.3	8.3	-8.2	8.2
Long. Pm (SUS)		1465.1	1465.1	1465.1	1465.1	1465.1	1465.1	1465.1	1465.1
Long. Pl (SUS)		-2.0	-2.0	-2.0	-2.0	-2.0	-2.0	-2.0	-2.0
Long. Q (SUS)		-2.3	2.3	-2.2	2.2	-2.3	2.3	-2.2	2.2
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.9	0.9	-0.9	-0.9	-0.9	-0.9	0.9	0.9
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		1465.1	1465.1	1465.1	1465.1	1465.1	1465.1	1465.1	1465.1
Pm+Pl (SUS)		1464.5	1464.5	1464.5	1464.5	1464.5	1464.5	1464.5	1464.5
Pm+Pl+Q (Total)		1461.0	1472.2	1461.1	1472.1	1461.0	1472.2	1461.1	1472.1

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1465.13	1509.10	Passed
Pm+Pl (SUS)	1464.48	2263.65	Passed
Pm+Pl+Q (TOTAL)	1472.25	4527.30	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: K250

From: 60

Pressure for Reinforcement Calculations	P	44.531	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	275	°C

Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²

Inside Diameter of Hemispherical Head	D	5389.00	mm
Head Finished (Minimum) Thickness	t	44.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	1500.0000	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		120.00	deg
Diameter		50.0000	mm

Size and Thickness Basis		Minimum	
Nominal Thickness		XXS	

Flange Material		SA-182 F11	
Flange Type		N/A	

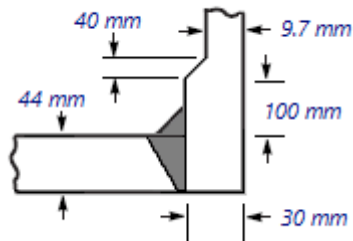
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	40.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm
Style of FVC Nozzle		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	44.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K250

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	40.945 mm.
Actual Thickness Used in Calculation	9.690 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 44.53 \cdot 2697.5 / (2 \cdot 1509 \cdot 1.0 - 0.2 \cdot 44.53)$
 $= 39.9171 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 44.53 \cdot 23.47 / (1407 \cdot 1.0 - 0.6 \cdot 44.53)$
 $= 0.7573 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3941 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 27.0$, less than the hub height, use (e-2)
 $= 67.5000 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

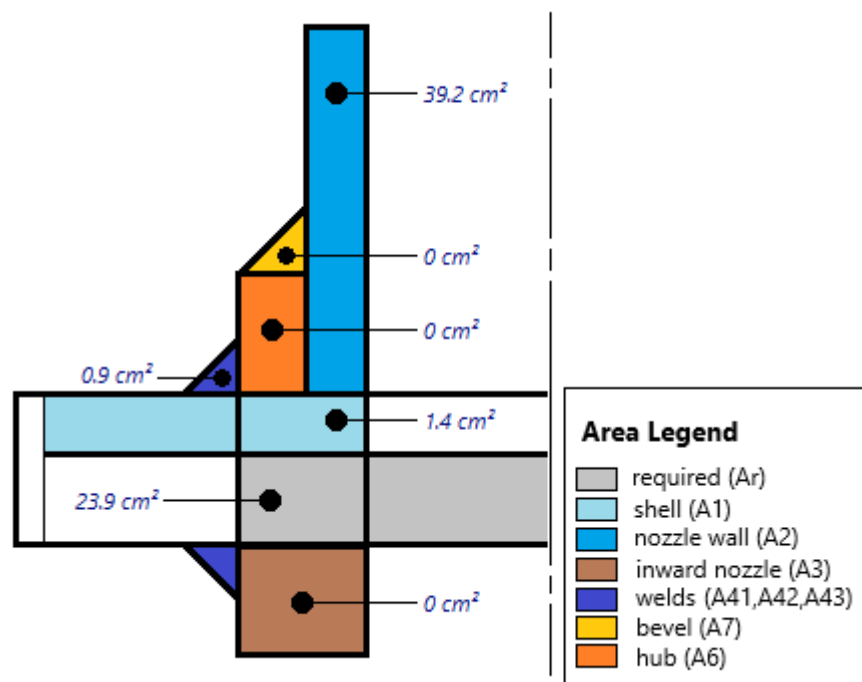
Parallel to Vessel Wall (Diameter Limit)	D1	192.3011 mm
Parallel to Vessel Wall	Rn+tn+t	96.1505 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	67.5000 mm

Hub Height was ≥ 2.5 times Hub Thickness, using sketch UG-40 (e-2).

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n/S_v)$
 $= \min(1, 1407.0/1509.1)$
 $= 0.932$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n/S_v)$
 $= \min(1, 1407.0/1509.1)$
 $= 0.932$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.932, 1.0)$
 $= 0.932$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	23.932	2.463	NA
Area in Shell	A1	1.433	43.388	NA
Area in Nozzle Wall	A2	39.220	39.763	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.932	0.932	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	41.585	84.083	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 56.49 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.3011 \cdot 39.9171 \cdot 1.0 + 2 \cdot 27.0 \cdot 39.9171 \cdot 1.0 \cdot (1 - 0.93))$
 $= 23.932 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 136.0(1.0 \cdot 41.0 - 1.0 \cdot 39.917) - 2 \cdot 27.0$
 $(1.0 \cdot 41.0 - 1.0 \cdot 39.9171) \cdot (1 - 0.932)$
 $= 1.433 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 67.5)(27.0 - 0.76)0.9323/\sin(57.4)$
 $= 39.220 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9323 + (0.0)^2 \cdot 0.9323$



= 0.932 cm²

There are no hub area calculations because Figure UG-40 (e-2) was used.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 9.69, tr = 0.757, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.113$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 30.0, tr = 0.757, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.028$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-26 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $44.53/52.72 = 0.845$

Weld Size Calculations, Description: K250

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= $\max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
= $\max(0, (23.9321 - 1.4332 + 2 \cdot 27.0 \cdot 0.9323 \cdot (1.0 \cdot 41.0 - 39.9171)) \cdot 1509)$
= 34775.89 kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= $(A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
= $(39.2199 + 0.9323 - 0.0 \cdot 0.93) \cdot 1509$
= 60593.66 kgf

Weld Load [W2]:
= $(A2 + A3 + A4 + (2 \cdot (\text{Hub Thickness}) \cdot t \cdot fr1)) \cdot Sv$
= $(39.2199 + 0.0 + 0.9323 + (20.6421)) \cdot 1509$
= 91744.59 kgf

Weld Load [W3]:
= $(A2+A3+A4+(2 \cdot (\text{Hub Thickness}) \cdot t \cdot fr1)) \cdot Sv$
= $(39.2199 + 0.0 + 0.9323 + 0.0) \cdot 1509$
= 91744.59 kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 121.0635 \cdot 10.0 \cdot 0.49 \cdot 1407 \\ &= 13111. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 44.3411) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 1407 \\ &= 37043. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 121.0635 \cdot (44.0 - 3.0) \cdot 0.74 \cdot 1509 \\ &= 87070. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (13111 + 37043) = 50154 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (13111 + 0 + 87070 + 0) = 100180 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (13111 + 87070 + 0) = 100180 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 50154	kgf, must exceed W = 34775 kgf	or W1 = 60593 kgf
Path 2-2 = 100180	kgf, must exceed W = 34775 kgf	or W2 = 91744 kgf
Path 3-3 = 100180	kgf, must exceed W = 34775 kgf	or W3 = 91744 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 46 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 kgf/cm²

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K250

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	40.945	mm.
Actual Thickness Used in Calculation	9.690	mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (2 \cdot Sv \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\ &= 44.53 \cdot 2697.5 / (2 \cdot 1509 \cdot 1.0 - 0.2 \cdot 44.53) \\ &= 39.9171 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned} &= P \cdot R / (Sn \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 44.53 \cdot 23.47 / (1407 \cdot 1.0 - 0.6 \cdot 44.53) \\ &= 0.7573 \text{ mm} \end{aligned}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.3941 mm

Intermediate Hub Nozzle Calculations:

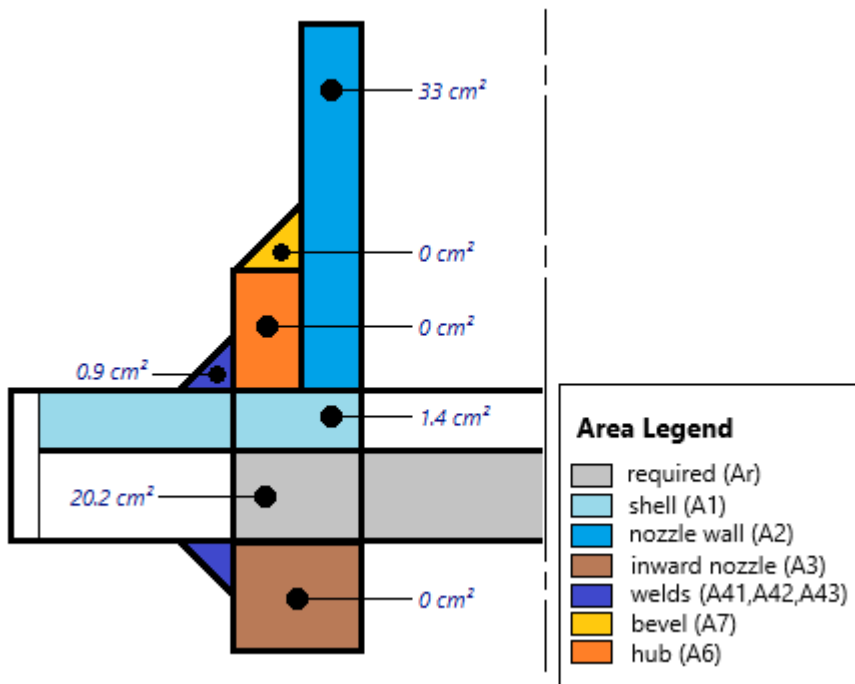
Check to determine use of Sketch (e-1) or (e-2):

$$\begin{aligned} &= 2.5 \cdot \text{Corroded Hub Thickness} \\ &= 2.5 \cdot 27.0, \text{ less than the hub height, use (e-2)} \\ &= 67.5000 \text{ mm} \end{aligned}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 182.9448 mm
 Parallel to Vessel Wall Rn+tn+t 91.4724 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 67.5000 mm

Hub Height was >= 2.5 times Hub Thickness, using sketch UG-40 (e-2).



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	20.197	2.079	NA
Area in Shell	A1	1.433	43.388	NA
Area in Nozzle Wall	A2	33.031	33.488	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.932	0.932	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	35.396	77.808	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (46.9448 \cdot 39.9171 \cdot 1.0 + 2 \cdot 27.0 \cdot 39.9171 \cdot 1.0 \cdot (1 - 0.93))$
 $= 20.197 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 136.0(1.0 \cdot 41.0 - 1.0 \cdot 39.917) - 2 \cdot 27.0$
 $(1.0 \cdot 41.0 - 1.0 \cdot 39.9171) \cdot (1 - 0.932)$
 $= 1.433 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$



$$= (2 \cdot 67.5)(27.0 - 0.76) 0.9323$$
$$= 33.031 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2$$
$$= 10.0^2 \cdot 0.9323 + (0.0)^2 \cdot 0.9323$$
$$= 0.932 \text{ cm}^2$$

There are no hub area calculations because Figure UG-40 (e-2) was used.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.7573 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 42.9171 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 42.9171 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.9426 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$
$$= \min[6.42, \max(42.9171, 4.5)]$$
$$= 6.4200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$
$$= \max(3.7573, 6.42)$$
$$= 6.4200 \text{ mm}$$

Available Nozzle Neck Thickness = 9.6901 mm -> OK

Weld Size Calculations, Description: K250

Intermediate Calc. for nozzle/shell welds	Tmin	19.0000 mm
---	------	------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot tn \cdot fr_1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$
$$= \max(0, (20.1974 - 1.4332 + 2 \cdot 27.0 \cdot 0.9323 \cdot (1.0 \cdot 41.0 - 39.9171)) \cdot 1509)$$
$$= 29139.79 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - Can/0.707)^2 \cdot fr_2) \cdot Sv$$
$$= (33.0308 + 0.9323 - 0.0 \cdot 0.93) \cdot 1509$$
$$= 51253.68 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot (\text{Hub Thickness}) \cdot t \cdot fr_1)) \cdot Sv$$
$$= (33.0308 + 0.0 + 0.9323 + (20.6421)) \cdot 1509$$
$$= 82404.61 \text{ kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + (2 \cdot (\text{Hub Thickness}) \cdot t \cdot fr_1)) \cdot Sv$$
$$= (33.0308 + 0.0 + 0.9323 + 0.0) \cdot 1509$$
$$= 82404.61 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_o \cdot W_o \cdot 0.49 \cdot Snw$$



$$= (3.1416/2.0) \cdot 100.9448 \cdot 10.0 \cdot 0.49 \cdot 1407$$
$$= 10932. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$
$$= (3.1416 \cdot 36.9724) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 1407$$
$$= 30888. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$
$$= (3.1416/2.0) \cdot 100.9448 \cdot (44.0 - 3.0) \cdot 0.74 \cdot 1509$$
$$= 72600. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (10932 + 30888) = 41819 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$
$$= (10932 + 0 + 72600 + 0) = 83532 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$
$$= (10932 + 72600 + 0) = 83532 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 41819	kgf, must exceed W = 29139 kgf	or W1 = 51253 kgf
Path 2-2 = 83531	kgf, must exceed W = 29139 kgf	or W2 = 82404 kgf
Path 3-3 = 83531	kgf, must exceed W = 29139 kgf	or W3 = 82404 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 46 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 kgf/cm²

The Drop for this Nozzle is : 34.6603 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 387.4367 mm

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Input, Nozzle Desc: K480

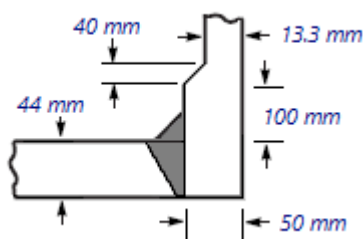
From: 60

Pressure for Reinforcement Calculations	P	44.531	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	5389.00	mm
Head Finished (Minimum) Thickness	t	44.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	2000.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		80.0000	mm
Size and Thickness Basis		Minimum	
Nominal Thickness		XXS	
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	40.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	2000.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	44.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.



Reinforcement CALCULATION, Description: K480

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	62.230 mm.
Actual Thickness Used in Calculation	13.335 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $44.53 \cdot 2697.5 / (2 \cdot 1509 \cdot 1.0 - 0.2 \cdot 44.53)$
= 39.9171 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $44.53 \cdot 34.12 / (1407 \cdot 1.0 - 0.6 \cdot 44.53)$
= 1.1006 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.8069 mm

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 36.665/0.5773
= 63.5057 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	203.4790 mm
Parallel to Vessel Wall	Rn+tn+t	101.7395 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	89.3432 mm

Weld Strength Reduction Factor [fr1]:

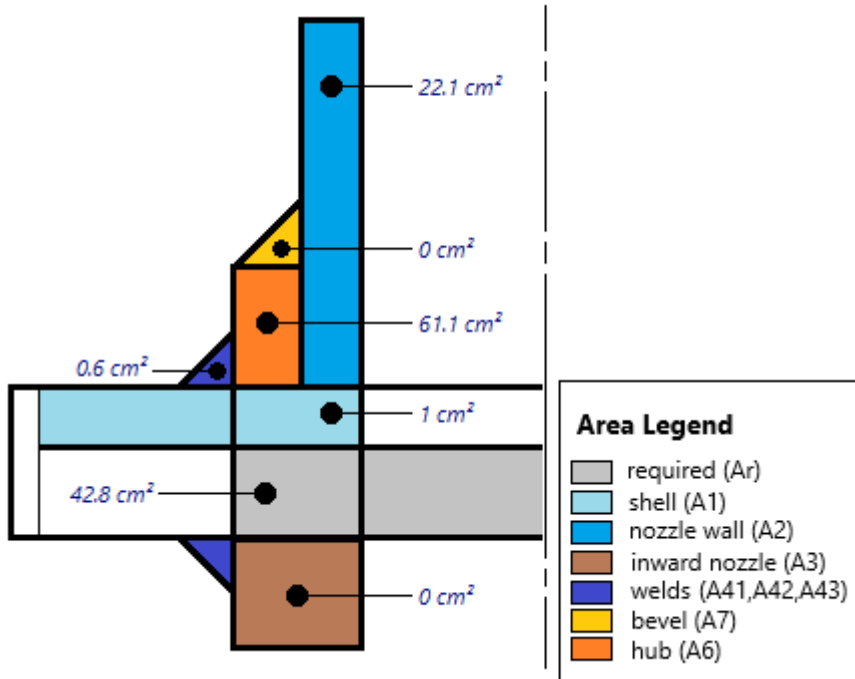
= min(1, Sn/Sv)
= min(1, 1407.0/1509.1)
= 0.932

Weld Strength Reduction Factor [fr2]:

= min(1, Sn/Sv)
= min(1, 1407.0/1509.1)
= 0.932

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)
= min(0.932, 1.0)
= 0.932



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	42.779	4.403	NA
Area in Shell	A1	1.043	31.574	NA
Area in Nozzle Wall	A2	22.098	22.801	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.633	0.633	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	61.083	61.083	NA
TOTAL AREA AVAILABLE	Atot	84.857	116.091	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 42.60 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (100.809 \cdot 39.9171 \cdot 1.0 + 2 \cdot 47.0 \cdot 39.9171 \cdot 1.0 \cdot (1 - 0.93))$
 $= 42.779 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 102.67(1.0 \cdot 41.0 - 1.0 \cdot 39.917) - 2 \cdot 47.0$
 $(1.0 \cdot 41.0 - 1.0 \cdot 39.9171) \cdot (1 - 0.932)$
 $= 1.043 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 89.34)(10.34 - 1.1)0.9323/\sin(44.1)$
 $= 22.098 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= (Wo^2 - \text{Area Lost}) \cdot fr2 + (Wi - can/0.707)^2 - \text{Area Lost}) \cdot fr2$
 $= (10.0^2 - 0.3209) \cdot 0.9323 + (0.0^2 - 0.0) \cdot 0.9323$



= 0.633 cm²

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2 \\ &= (2 \cdot \min(89.3, 2000.0, 100.0)) \cdot (50.0 - 13.3) \cdot 0.932 \\ &= 61.083 \text{ cm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 13.335, tr = 1.101, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.106, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-20 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 44.0, tr = 39.917, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.974, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	33 °C
Min Metal Temp. at Required thickness (UCS 66.1)	32 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-26 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 44.53/52.72 = 0.845

Weld Size Calculations, Description: K480

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A-A1+2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv) \\ &= \max(0, (42.7787 - 1.0429 + 2 \cdot 47.0 \cdot 0.9323 \cdot (1.0 \cdot 41.0 - 39.9171)) \cdot 1509) \\ &= 64415.54 \text{ kgf} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A4-(W_i \cdot Can/.707)^2 \cdot fr2) \cdot Sv \\ &= (83.1809 + 0.6331 - 0.0 \cdot 0.93) \cdot 1509 \\ &= 126483.59 \text{ kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv \\ &= (83.1809 + 0.0 + 0.6331 + (35.9325)) \cdot 1509 \\ &= 180709.30 \text{ kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S \\ &= (83.1809 + 0.0 + 0.6331 + 0.0 + (35.9325)) \cdot 1509 \\ &= 180709.30 \text{ kgf} \end{aligned}$$



Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 239.6929 \cdot 10.0 \cdot 0.49 \cdot 1407 \\ &= 25958. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 85.1255) \cdot (50.0 - 3.0) \cdot 0.7 \cdot 1407 \\ &= 123794. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 239.6929 \cdot (44.0 - 3.0) \cdot 0.74 \cdot 1509 \\ &= 172388. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (25958 + 123794) = 149751 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (25958 + 0 + 172388 + 0) = 198346 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (25958 + 172388 + 0) = 198346 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 149751 kgf, must exceed W = 64415 kgf	or W1 = 126483 kgf
Path 2-2 = 198346 kgf, must exceed W = 64415 kgf	or W2 = 180709 kgf
Path 3-3 = 198346 kgf, must exceed W = 64415 kgf	or W3 = 180709 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 46 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 kgf/cm²

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K480

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	62.230 mm.
Actual Thickness Used in Calculation	13.335 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\ &= 44.53 \cdot 2697.5 / (2 \cdot 1509 \cdot 1.0 - 0.2 \cdot 44.53) \\ &= 39.9171 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned} &= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 44.53 \cdot 34.12 / (1407 \cdot 1.0 - 0.6 \cdot 44.53) \\ &= 1.1006 \text{ mm} \end{aligned}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.8069 mm

Intermediate Hub Nozzle Calculations:

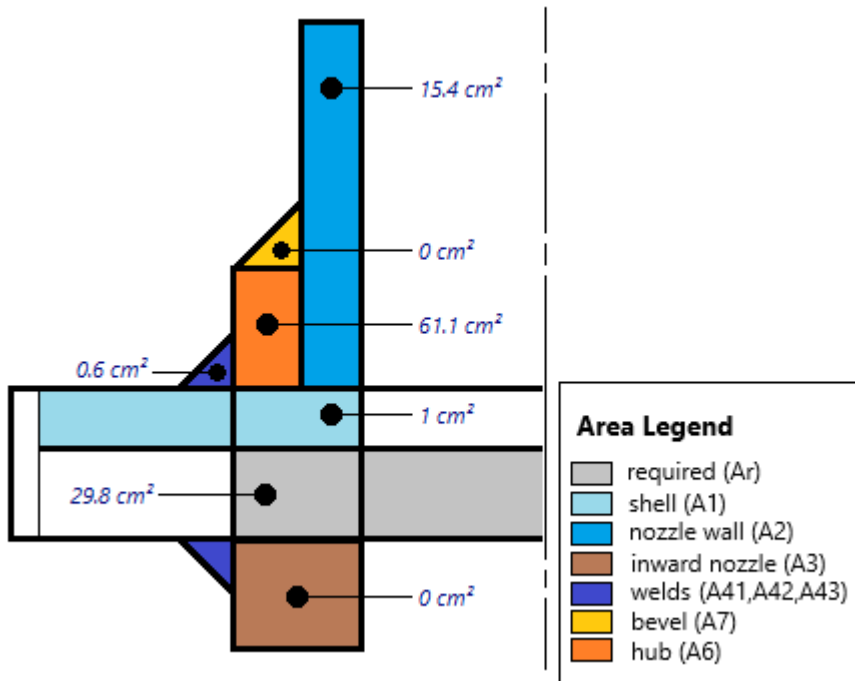
Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 36.665 / 0.5773$
 $= 63.5057 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 170.9000 mm
 Parallel to Vessel Wall Rn+tn+t 85.4500 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 89.3432 mm



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	29.774	3.064	NA
Area in Shell	A1	1.043	31.574	NA
Area in Nozzle Wall	A2	15.384	15.874	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.633	0.633	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	61.083	61.083	NA
TOTAL AREA AVAILABLE	Atot	78.143	109.164	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (68.23 \cdot 39.9171 \cdot 1.0 + 2 \cdot 47.0 \cdot 39.9171 \cdot 1.0 \cdot (1 - 0.93))$
 $= 29.774 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 102.67(1.0 \cdot 41.0 - 1.0 \cdot 39.917) - 2 \cdot 47.0$



$$(1.0 \cdot 41.0 - 1.0 \cdot 39.9171) \cdot (1 - 0.932) \\ = 1.043 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_r) f_{r2} \\ = (2 \cdot 89.34) (10.34 - 1.1) 0.9323 \\ = 15.384 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (W_o^2 - \text{Area Lost}) \cdot f_{r2} + (W_i - \text{can}/0.707)^2 - \text{Area Lost} \cdot f_{r2} \\ = (10.0^2 - 0.3209) \cdot 0.9323 + (0.0^2 - 0.0) \cdot 0.9323 \\ = 0.633 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2} \\ = (2 \cdot \min(89.3, 2000.0, 100.0)) \cdot (50.0 - 13.3) \cdot 0.932 \\ = 61.083 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.1006 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 42.9171 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 42.9171 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.9426 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)] \\ = \min[7.8, \max(42.9171, 4.5)] \\ = 7.8000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb) \\ = \max(4.1006, 7.8) \\ = 7.8000 \text{ mm}$$

Available Nozzle Neck Thickness = 13.3350 mm -> OK

Weld Size Calculations, Description: K480

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) S_v) \\ = \max(0, (29.7741 - 1.0429 + 2 \cdot 47.0 \cdot 0.9323 \cdot (1.0 \cdot 41.0 - 39.9171)) 1509) \\ = 44790.31 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\ = (76.467 + 0.6331 - 0.0 \cdot 0.93) \cdot 1509 \\ = 116351.69 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\ = (76.467 + 0.0 + 0.6331 + (35.9325)) \cdot 1509 \\ = 170577.39 \text{ kgf}$$

Weld Load [W3]:



$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r 1)) \cdot S \\ &= (76.467 + 0.0 + 0.6331 + 0.0 + (35.9325)) \cdot 1509 \\ &= 170577.39 \text{ kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 162.23 \cdot 10.0 \cdot 0.49 \cdot 1407 \\ &= 17569. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 57.615) \cdot (50.0 - 3.0) \cdot 0.7 \cdot 1407 \\ &= 83787. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 162.23 \cdot (44.0 - 3.0) \cdot 0.74 \cdot 1509 \\ &= 116677. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (17569 + 83787) = 101355 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (17569 + 0 + 116677 + 0) = 134245 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (17569 + 116677 + 0) = 134245 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 101355 kgf, must exceed W = 44790 kgf	or W1 = 116351 kgf
Path 2-2 = 134245 kgf, must exceed W = 44790 kgf	or W2 = 170577 kgf
Path 3-3 = 134245 kgf, must exceed W = 44790 kgf	or W3 = 170577 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 46 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 kgf/cm²

The Drop for this Nozzle is : 94.1214 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 2159.1465 mm

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